



Cold start of proton exchange membrane fuel cell



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ABSTRACT

In this review, “cold start” is defined as the startup of proton exchange membrane (PEM) fuel cells from sub-freezing temperatures. Problems occurring during the cold start pose some of the remaining barriers to commercial applications of PEM fuel cells in transportation, stationary, auxiliary and portable systems. Fundamental studies of transport phenomena are critical to a better understanding of the mechanisms of cold start and offer ultimate solutions to resolving cold-start issues. In this review, experimental studies are discussed, focusing on output performance degradation, water and ice visualization, and component damages during a cold start. Analytical, numerical, and microscopic models and their results are also discussed. One of the emphases is on transport phenomena relevant to cold starts, including supercooling, phase change and transport of water in the membrane, catalyst layer, microporous layer, and gas diffusion layer. Another emphasis is placed on the strategies utilized to optimize cold-start processes for improved performance. The strategies include material designs of the components, cell/stack structures, and startup mode/load controls. It is shown that all of the effective strategies to mitigating cold-start problems derive from a basic understanding of the transport mechanisms during a cold start. It is also suggested that future models for this problem should place a great deal of attention in supercooling phenomena and water phase-change and transport in multilayer porous media. Lastly, more advanced experimental methods, such as real-time water/ice visualization and cryogenic microscopy, are needed to validate emerging theories and models.

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Contents

1. Introduction	30
1.1. Origin and importance of cold start	30
1.2. Technical targets	32
1.3. State-of-the-art	32
1.4. Scope and objectives	32
2. Experiments	32
2.1. Output performance	33
2.2. Visualization of liquid and ice	36
2.3. Freeze damage and degradation	37
2.4. Cold start mechanisms	37
2.5. Summary	37
3. Overview of models	38
3.1. Analytical model	38
3.2. Numerical model	40
3.2.1. One-dimensional numerical model	41
3.2.2. Multi-dimensional multiphase model	42
3.3. Microscopic model	43
3.4. Model validation	43
3.5. Summary	44
4. Transport phenomena in cold start	45
4.1. Water transport	45

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4.1.1.	Membrane processes	45
4.1.2.	Porous layers	45
4.1.2.1.	Ionomer water transport in CL	45
4.1.2.2.	General descriptions of water phase change in CL	45
4.1.2.3.	Origin of the supercooling phenomenon	45
4.1.2.4.	Effects of cold-start factors on the supercooling phenomenon	46
4.1.2.5.	Models of supercooling phenomenon	46
4.1.2.6.	Suggestions on future supercooling modeling	46
4.1.2.7.	Ice formation in GDL	47
4.1.2.8.	The microporous layer	47
4.1.2.9.	Effects of MPL on liquid water transport	47
4.1.2.10.	Effects of MPL on cold start performance	48
4.1.3.	Flow channels	49
4.2.	Gas species transport	49
4.2.1.	Flow channels	49
4.2.2.	Porous layers	49
4.3.	Heat transfer and heat sources	50
4.4.	Charge transport	50
4.5.	Summary	50
5.	Optimizations	50
5.1.	Design optimization	51
5.1.1.	Membrane	51
5.1.2.	Catalyst layer	51
5.1.3.	Microporous layer	51
5.1.4.	Gas diffusion layer	51
5.1.5.	Flow channels	52
5.1.6.	Stack	52
5.2.	Load control	52
5.2.1.	Current	52
5.2.2.	Voltage	52
5.2.3.	Power	52
5.3.	Assisted start-up	53
5.3.1.	Purging	53
5.3.2.	Heating	54
5.3.2.1.	Electric heating	54
5.3.2.2.	Gas heating	54
5.3.2.3.	Coolant heating	55
5.3.2.4.	Internal catalytic hydrogen-oxygen reaction	55
5.3.2.4.	External burner	55
5.4.	Summary	55
6.	Challenges and opportunities	56
6.1.	Existing challenges and expected improvements	56
6.1.1.	Liquid-ice transition theory	56
6.1.2.	Multi-scale modeling	56
6.1.3.	Gap between required performance and existing technologies	57
6.2.	Opportunities	57
6.2.1.	Experimental studies	57
6.2.2.	Modeling studies	57
6.2.3.	Model uncertainty	57
6.2.4.	Material development	57
7.	Concluding remarks	57
	Supplementary materials	58

1. Introduction

Proton exchange membrane fuel cells (PEMFCs) are a promising energy conversion device that has the potential for applications in automotive, stationary, portable, auxiliary power systems, and even submarines and space shuttles [1]. A PEM fuel cell consumes hydrogen and oxygen in the anode and cathode electrodes, respectively. It converts chemical energy directly into electrical energy. PEM fuel cells have unique advantages over traditional combustion engines. As a power source, it is clean, since the only exhaust is water. It works quietly and gently, as there are no moving components inside. It can have high power density and operate at low temperatures [2].

1.1. Origin and importance of cold start

One of the most exciting prospects of PEM fuel cells is in automotive applications. In 2017, three major automakers (Toyota, Hyundai, and Honda) will release their latest fuel cell vehicles (FCVs) to the market [3]. FCVs are expected to survive in a variety of environments and especially under extreme conditions, such as subfreezing temperatures. The normal operation of PEM fuel cells requires a constant supply of fuel and oxidants from the flow channel to the catalyst layer, as well as the drainage of produced water. In subfreezing temperatures, the water produced can freeze in porous layers and even in flow channels. Ice then accumulates, blocking gas transport

Nomenclature

a	water activity
A	stack geometric area, reaction area, m^2
ASR	area specific resistance, $\Omega \cdot \text{cm}^2$
c	molar concentration, mol m^{-3}
C_p	specific heat, $\text{J kg}^{-1} \text{K}^{-1}$
D	mass diffusivity, $\text{m}^2 \text{s}^{-1}$
D_h	hydraulic diameter, m
ECA	electrochemical active area
EW	equivalent weight of membrane, $1100 \text{ kg kmol}^{-1}$
f	friction factor
F	Faraday's constant, $96,487 \text{ C mol}^{-1}$
G	Gibbs free energy, J
h	latent heat, J kg^{-1}
h_0	heat transfer coefficient, $\text{W m}^{-2} \text{K}^{-1}$
h_{pc}	enthalpy change, J mol^{-1}
H_f	heat of fusion, J kg^{-1}
I	current density, A cm^{-2}
j	reaction rate, A m^{-3}
j^*	exchange current density, A cm^{-2}
J	nucleation rate, nuclei $\text{m}^{-3} \text{s}^{-1}$
k	thermal conductivity, $\text{W m}^{-1} \text{K}^{-1}$
K	permeability, m^2
L	length, m
m	mass, kg
$\dot{m}$	mass flow/transfer rate, kg s^{-1}
m^{therm}	thermal mass
M	molecular weight, kg kmol^{-1}
MEA	membrane electrode assembly
n^{H_2O}	water storage capacity, mol
$\dot{n}^{H_2O}$	water generation/removal rate, mol s^{-1}
n_d	electro-osmotic drag coefficient
N^K	Knudsen diffusion flux, $\text{mol m}^{-2} \text{s}^{-1}$
p	pressure, Pa
p_c	capillary pressure, Pa
q	heat flux, W m^{-2}
$\dot{Q}$	heat transfer rate, W
r	nucleus radius, m
r^*	critical cluster radius, m
R	universal gas constant, $\text{J mol}^{-1} \text{K}^{-1}$
RH	relative humidity
s	volume fraction
S	source terms, entropy, $\text{J kmol}^{-1} \text{K}^{-1}$
S_{lq}	liquid saturation
ΔS	entropy change
t	time, s
T	temperature, K or $^\circ\text{C}$
T^0	standard temperature, 298 K
T_g	melting temperature, K
u	velocity, m s^{-1}
U_0	reference open circuit potential, V
U_{cell}	cell voltage, V
v	molar volume, $\text{m}^3 \text{mol}^{-1}$
V	voltage
Y	mass fraction

Greek letters

ε	porosity
α	transfer coefficient
β	heating rate, K s^{-1}
δ	thickness, m
ε_e	volume fraction of ionomer

ζ	water transfer rate, s^{-1}
ζ^K	correction factor for Knudsen diffusion coefficient
η	over potential, V
θ	contact angle, $^\circ$
ι	interfacial drag coefficient
κ	electrical conductivity, S m^{-1}
λ	water content in ionomer
μ	dynamic viscosity, $\text{kg m}^{-1} \text{s}^{-1}$
ξ	stoichiometry ratio
ξ_i	bend loss coefficient
ρ	density, kg m^{-3}
σ	surface tension, N m^{-1}
σ_{cl}	interfacial energy between cluster and water, J cm^{-2}
τ_d	ice blockage coefficient for gas diffusivity
τ_a	ice coverage coefficient
ϕ	electrical potential, V
ϕ_{ice}	volume fraction of water which transfers to ice
ω	volume fraction of ionomer in catalyst layer

Subscripts and superscripts

0	initial condition
a	anode
act	activation
acc	accumulation
atm	atmosphere
BP	bipolar plate
c	cathode
$cell$	cell characteristic
$channel$	flow channel
$conc$	concentration
dry	dry
D	diffusion
eff	effective
env	environment
eq	equilibrium
ele	electronic
EOD	electro-osmotic drag
f	frozen
fl	fluid phase
fmw	frozen membrane water
g	gas phase
gen	generation
GDL	gas diffusion layer
H_2	hydrogen
H_2O	water
i	the i th component or the i th cell in a stack
ice	ice
in	inlet
$init$	initial condition
ion	ionic
$irrev$	irreversible
lq	liquid water
$loss$	loss
m	mass (for source term)
mem	membrane
$melt$	melt
MPL	microporous layer
$nernst$	Nernst
nf	non-frozen
nmw	non-frozen membrane water

<i>N</i>	total number of cells in a stack
<i>out</i>	outlet
<i>ohmic</i>	ohmic
<i>O₂</i>	oxygen
<i>pc</i>	phase change
<i>P</i>	power
<i>ref</i>	reference state
<i>rev</i>	reversible
<i>sat</i>	saturation
<i>sl</i>	solid phase
<i>stk</i>	stack characteristic
<i>surr</i>	surroundings
<i>stack</i>	stack characteristic
<i>t</i>	time
<i>T</i>	energy (for source term)
<i>u</i>	momentum (for source term)
<i>vp</i>	water vapor
<i>w</i>	water
<i>wall</i>	surrounding wall of the single cell or stack
<i>l-i</i>	liquid water to ice (vice versa)
<i>n-f</i>	non-frozen membrane water to frozen membrane water (vice versa)
<i>n-i</i>	non-frozen membrane water to ice
<i>n-v</i>	non-frozen membrane water to vapor (vice versa)
<i>v-i</i>	vapor to ice
<i>v-l</i>	vapor to water liquid (vice versa)

pathways and covering catalyst reaction sites, which eventually causes start-up failures. It is known that oxygen reduction at sub-freezing temperatures is as facile as that under ambient temperature conditions [4]. Therefore, the major factors governing cold-start problems are those related to the various transport phenomena inside PEM fuel cells.

Fig. 1 presents the major transport phenomena in PEM fuel cells during a cold start. Components of a PEM fuel cell include the membrane, catalyst layer (CL), gas diffusion layer (GDL), micro-porous layer (MPL) and bipolar plate (BP). Hydrogen and air flows pass through the anode and cathode flow channels, respectively. Diffusion and convection of the gases co-exist in the porous layers (GDL, MPL, and CL). The CL is comprised of a mixture of catalyst particles, ionomer, and porous carbon backbone. Electrochemical reactions occur on the three-phase coexistence sites (ionomer, gas, and catalyst) in the CL. Electricity is generated during operation, along with water as the reaction product. During a cold start, water transforms from one phase or state to another. It can be absorbed by the ionomer and become membrane water. Part of the membrane water can transform to frozen membrane water due to the subfreezing temperature [5]. Water can also evaporate from the ionomer. The resulting vapor percolates through the porous layers and enters into the flow channel. Water vapor can also deposit and accumulate in the porous layers as ice. Lastly, water can stay in a supercooled liquid state under certain conditions [6]. During a cold start, temperature rises due to the exothermic electrochemical reaction. A successful cold start requires that the CL temperature exceeds the ice's melting point before the reaction sites and diffusion pathways are blocked. In this case, ice melts and liquid water can be drained through the flow channel.

1.2. Technical targets

The cold start ability of PEM fuel cells is one of the critical targets in many existing fuel cell technical roadmaps. In 2005, the US Department of Energy (DOE) set the first cold start target: to achieve a successful cold start from -20°C and up before 2010 [7]. This target temperature was further lowered in 2013. The 2017–2020 DOE fuel cell plan calls for an assisted cold start from -30°C and rapid startup from -20°C within 30 s to reach 50% of rated power [8]. The latest targets released by the DOE include an unassisted start from -30°C and assisted start from -40°C , both requiring an 8 h soak at the indicated temperatures [9]. The cold start target in Europe is -25°C for fuel cell development scenarios from 2015 to 2020 [10].

1.3. State-of-the-art

The first practical hydrogen-oxygen fuel cell stack was demonstrated in the 1960s [2]. Decades of research and development have led to significant progress in the cold start performance of PEM fuel cells. Such performance can be evaluated by two factors: the lower limit of startup temperature, and startup time, as shown in Fig. 2. Startup time is defined as the time to reach 50% of rated power. For automotive applications, cold start performance has improved significantly in recent years. Honda delivered its first FCX fuel cell cars in 2002. It achieved a cold start from -11°C in 2004 [11]. It was reported that in 2005, Hyundai's Tucson FCEV was capable of starting up from -10°C [12]. Ballard achieved a cold start temperature of -20°C in 2004 [13]. In 2009, the cold start limit was further improved to -30°C by the Toyota FCHV-adv, and it endured temperatures as low as -37°C during operation [14]. Toyota's Mirai was reported to have a cold start temperature of -30°C [15].

1.4. Scope and objectives

The main purpose of this article is to provide a comprehensive review of the experimental and modeling studies of transport phenomena during the cold start of PEM fuel cells. Suggestions are made for improvements of theories governing the various cold start phenomena. Optimization strategies for cold start performance are also summarized and evaluated. This article is organized as follows: Sections 2 and 3 present a review of experiments and modeling studies on cold start issues, respectively; Section 4 presents discussions on transport phenomena; Section 5 summarizes optimization strategies aiming to improve cold start performances; and Section 6 discusses challenges and research opportunities. Finally, concluding remarks are given in Section 7.

2. Experiments

Observations in experimental studies are the foundation for theory development and cold start performance optimization. In one regard, experiments are capable of revealing new phenomena with the development of characterization techniques. On the other hand, the validation of existing theories and models require systematic experimentation. Validated theories and accurate models can lead to better predictions and optimizations of cold start performances in a practical aspect for FCVs. As illustrated in Fig. 3, experimental studies can be categorized into three aspects. First, tests of output performance present correlations among various cold start conditions and startup performances. Secondly, visualizations of water and ice in cold starts provide insight into the mechanisms of performance deterioration. Third, analyses of damage due to component freezing contribute to a better understanding of cold start effects on fuel cell durability. Fig. 3 only illustrates representative experimental results. Detailed literature surveys on experimental studies of cold start are summarized in Table S1 [6, 16–41] in Supplementary Materials. The

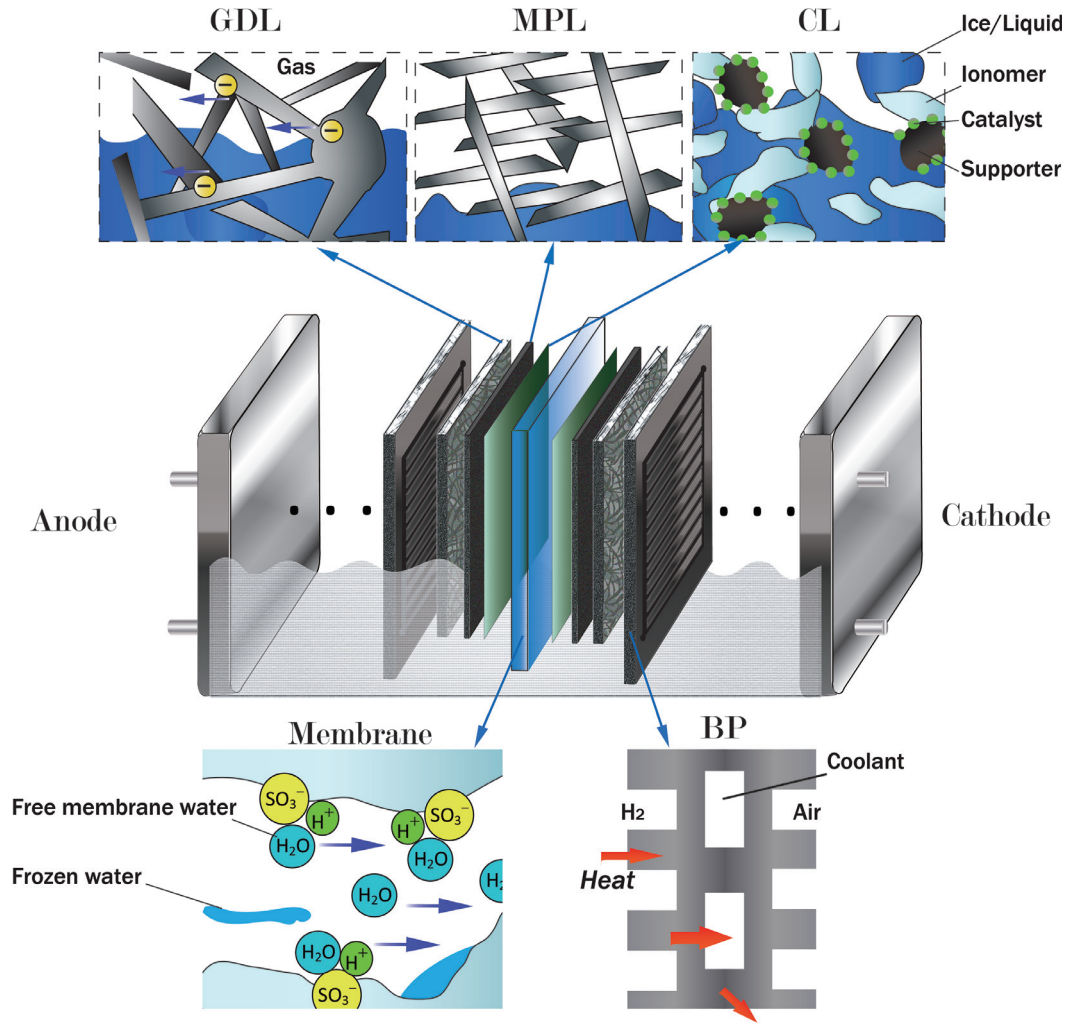


Fig. 1. Schematic diagram of major transport phenomena in the cold start of the proton exchange membrane fuel cell.

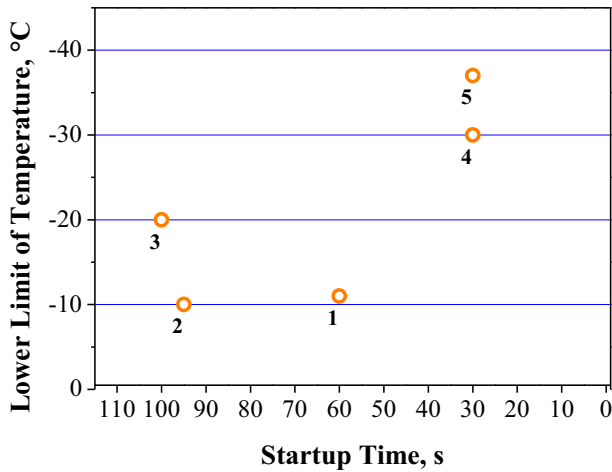


Fig. 2. PEMFC cold start performance as reported in the literature. Y-axis denotes the lower limit of start-up temperature, and X-axis is the corresponding start-up time. 1. Honda FCX, 2004 [11]; 2. Hyundai Tucson FCEV, 2005 [12]; 3. Ballard, 2004 [13]; 4. Toyota, Mirai, 2016 [15]; 5. Toyota FCHV-adv, 2008 [14].

following subsections discuss the aforementioned three aspects of cold start experiments.

2.1. Output performance

In the first level of experiments, cold start processes are characterized by measuring the output performances; i.e. current density, voltage, and temperature [16–18,25,27,42–50]. Cell voltage and current density are major performance factors, as they determine power output. Cell temperature is critical to the cold-start process. Temperature rise is an essential requirement for a successful startup. A testing workbench for cold-start output typically includes a single PEM fuel cell or stack, a test station, a gas feeding system, and measuring instruments, such as thermocouples and an environmental chamber which maintains controlled temperatures [25]. The test station controls cell current density, voltage and gas feeding conditions. The gas feeding system supplies hydrogen and air as reactants. It also supplies nitrogen in a purging operation. Thermocouples are usually embedded in the flow field plates. The measured temperature may be lower than the actual temperature of the membrane

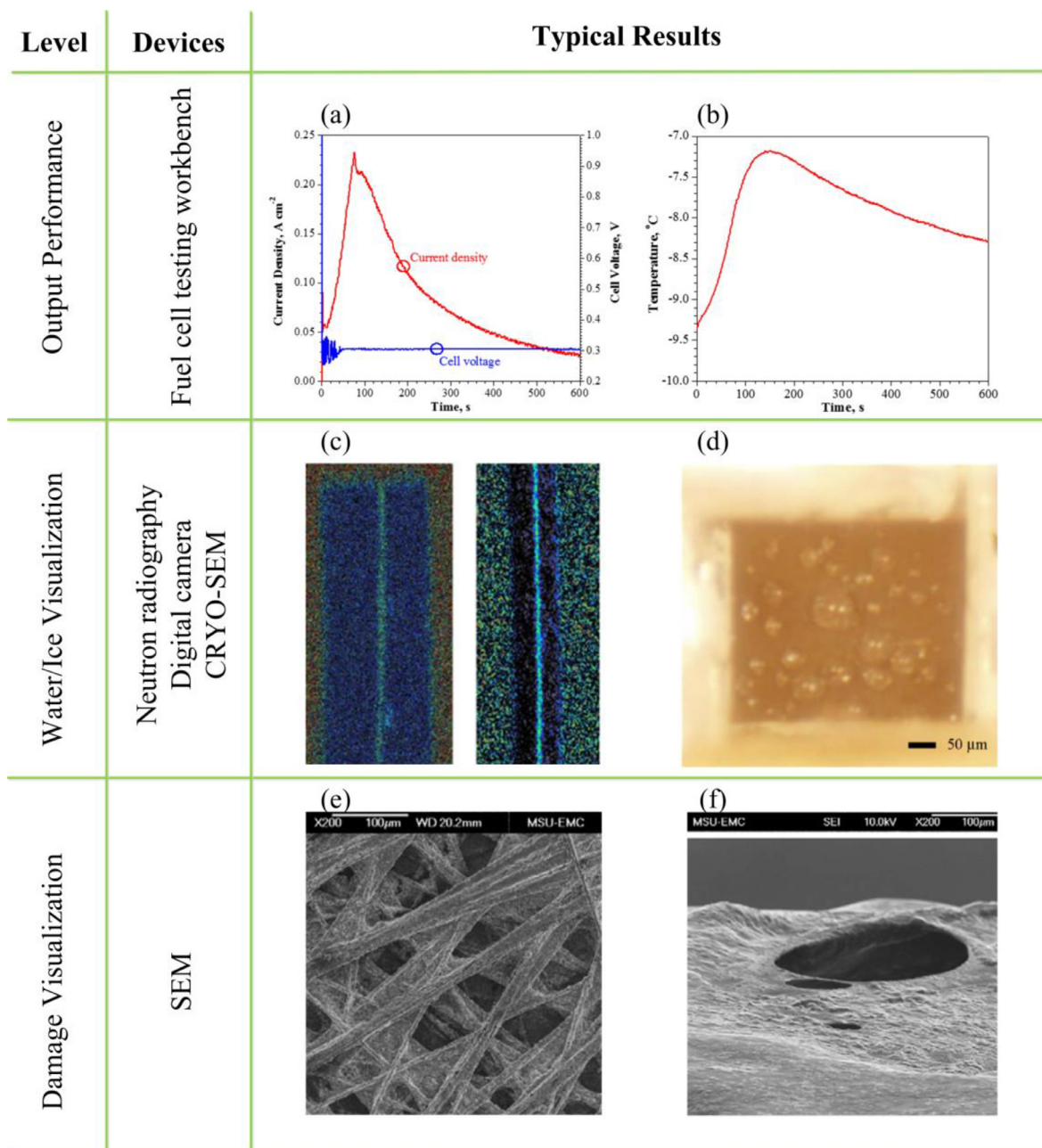


Fig. 3. Three types of experimental studies on the PEMFC cold start. (a), (b): evolution of current density, cell voltage, and cell temperature during cold start (adapted from [25]). (c): visualization of water and ice in cross-section through-plane (adapted from [30]); (d): visualization of water and ice on MEA surface (adapted from [31]). (e), (f): microscopy image of GDL and membrane after cold-start cycles (adapted from [16]).

electrode assembly (MEA). The environment chamber can be made with insulation materials. It is cooled down with the circulating coolant. During cold start experiments, fuel cells dissipate heat to the simulated subfreezing environment. It should be noted that the actual temperature inside the chamber may vary.

During a constant-voltage cold start process as shown in Fig. 3(a) and (b), initially, the voltage has some fluctuation. It indicates that temperature is low and cell resistance is high in the initial state, which results in unstable voltage control [25,27]. Current density then either increases or decreases, according to the initial state of the membrane. When the ionomer is initially dry, the current density is more likely to increase in the early stage because the dry membrane is hydrated by produced water, thus increasing conductivity [21]. The current-decrease situation occurs when initial water content and current density are both high during the early stage.

Strong electro-osmotic drag (EOD) effect drives water to move from the anode to cathode in MEA. Resistance in the anode increases significantly due to the lack of water. As a result, the current density decreases [18]. In the next stage, current density decreases slightly; because ice content increases, the effects of ice blockage on the reaction sites and reactant supply pathways become more notable and consequently suppress the effects of ionomer hydration. In the final stage, cold start failure occurs. Based on the observations of output performance in experiments, there are several factors that affect various cold start failures. These include the initial temperature, operational current density, and the presence of supercooled water.

Temperature has a varied effect upon the nature of cold start failure. When the initial temperature is low, such as at -10°C , the failure process is relatively gentle, as shown in Fig. 3(a) and (b). If the initial temperature is slightly higher, around -7°C , failure occurs

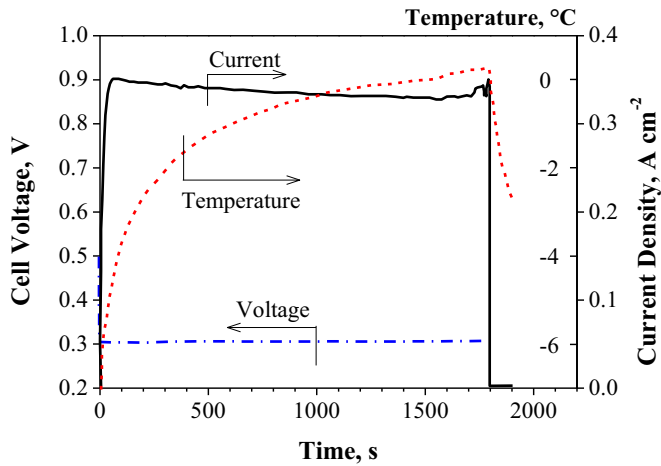


Fig. 4. Evolution of current density, cell voltage, and cell temperature during a cold start from -7°C (adapted from [25]).

suddenly after a period of steady operation [25]. Fig. 4 shows such a case in constant voltage mode (at 0.3 V). In the initial stage, current density increases quickly and then remains steady, indicating that the electrochemical reaction is not greatly hindered by ice. However, the cell suddenly stops working at around 1790 s as the current density drops to zero and the cell temperature ceases to increase. The fact that a slight change in the initial temperature can result in different failure mechanisms indicates that there is an initial temperature threshold. For the initial temperature at -10°C , the temperature is not increased quickly enough to allow the CL to reach melting point, and ice is gradually formed in that layer. The ice grows to cover the reaction sites in the CL, leading to a gradual decrease of output current density. In the cold start case from -7°C , the temperature inside the CL exceeds the melting point before ice fully blocks the CL, even as the flow channel is still at subfreezing temperatures. Liquid water freezes in it and subsequently blocks the gas supply. This explanation is supported by the sudden pressure drop in the cathode flow channel.

As for the effects of operational current density, a cold start scenario with higher current density fails much faster than one with lower current density, as shown in Fig. 5 [43]. The cumulative amount of water product can be calculated from the integral of current density over time. In the two cases shown in Fig. 5, the cumulative water amounts are 0.32 and 0.53 mg/cm^2 at the shutdown point for the high and low current case, respectively. This result indicates

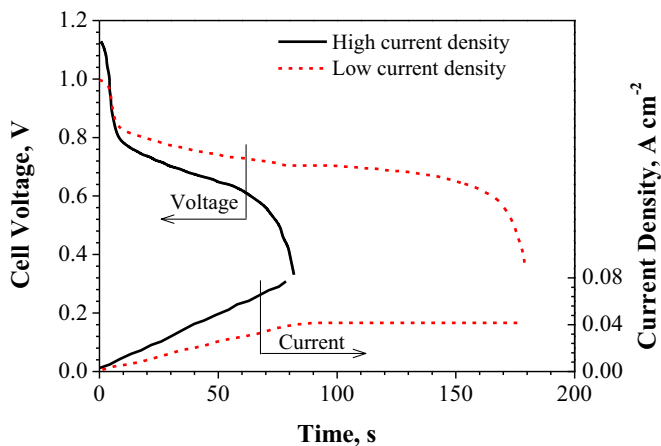


Fig. 5. Evolution of current density and cell voltage in two cold starts with different operational current densities from -30°C (adapted from [43]).

that the capacity of the CL for ice storage is not fully utilized in the case of a high current. The mechanism of rapid failure with high current density is that ice distribution in the CL is not uniform with a high current density. Because water generation rate is high in this situation, there is not enough time for water to redistribute uniformly in the CL. Therefore, water tends to concentrate on the GDL-side of the CL, causing ice to quickly obstruct this region. As a result, gas can no longer be fed into the CL, resulting in cold-start failure.

Thirdly, the presence of supercooled water could lead to a sudden failure of the cold start [21]. If supercooled water is absent, the cold start is more likely to fail gradually. For these two different failure modes, when supercooled water is present, liquid state water moves out of the CL and GDL and into the flow channel, thus preventing blockage of the CL. Once the release of supercooled water is triggered, it rapidly becomes ice. Subsequently, the gas supply pathways and reaction sites are suddenly blocked by ice and leads to sudden failure. The supercooling phenomenon is supported by micrography observations of supercooled water. It is discussed in detail in the next subsection.

While the discussions above focus on uncovering cold start failure modes and mechanisms, other experimental studies aim to figure out the limitations of cold start ability. For single cells, the minimum successful startup temperature ranges from -3°C to -7°C , depending on different environment and operation conditions. For example, in [26], it is found that the single cell can successfully start up from -7°C with a current density of 0.04 A cm^{-2} and 0.08 A cm^{-2} , although it fails with 0.12 A cm^{-2} . With an initial temperature of -3°C , no failure as observed over current densities from 0.04 to 0.12 A cm^{-2} . In [16], it is found that a cold start from -5°C is possible without causing degradation. Pinton et al. [18] also concluded that a cold start from -5°C could be achieved. Jiao et al. [27] showed that their single cell could successfully start from -5°C , though it failed at -10°C . Lowering the unassisted cold start temperature is the objective of cell design, material, and operation optimization. This subject is discussed further in Section 5.

Most of the cold start experiments were conducted based on single cells, necessitating experiments on stacks and systems; cold start characteristics may be markedly different when single cells are integrated into stacks. Stacks have lower thermal mass than experimental single cells. Therefore, temperature variation in stacks is expected to be larger than that in single cells. Furthermore, optimizations of cold-start performance usually requires consideration at the system level.

A cold-start experiment was conducted using a portable 6-cell stack with a 30 cm^2 active area [22]. It was found that a successful passive cold start could be achieved from -10°C , but it was not possible to start up from -20°C . Jia et al. [20] experimentally studied the cold start dynamic performance of a 2-cell stack. The effects of various conditions were tested, including the ambient temperature, operation current density, and stoichiometry. It was found that the 2-cell stack could start up without requiring extra energy at -5°C . The different initial water contents were also compared, indicating that higher water content may be beneficial to the cold start performance. Han et al. [51] used a 5-cell stack to conduct the cold start experiments to explore the effects of GDL anisotropic bending stiffness. The GDL materials were installed in the stack in two different directions: one with the higher stiffness direction aligned with the flow field direction (identified as 0° GDL), and the other placed perpendicular to the flow direction (identified as 90° GDL). The experiment found that the I-V performance of the 90° GDL stack was higher than that of the 0° GDL stack after cold start cycles.

A PEM fuel cell system was developed for portable applications [23]. The core of this system was a 6-cell stack with an active surface of 30 cm^2 . Aside from basic thermal and water management auxiliary equipment, an electric heating device was attached to assist the cold start. In that study, the cold start processes from -20°C were

conducted with external heating of 30 W. It was found that a successful cold start can be achieved from -20°C , and long term operation in ambient -20°C was stable. This study confirmed the importance of a complete system design for a PEM fuel cell cold start. Schießwohl et al. [24] set up a PEM fuel cell system to experimentally study the critical factors to improving cold start ability. The shutdown strategy has strong effects on the cold start time. It is concluded that the minimum successful cold start temperature of a certain system mainly depends on the thermal mass of the system, and system thermal mass mainly depends upon the cooling loops. If coolant does not cycle during the cold start, only 200 s is needed to reach 50% power in the system. It indicates that deactivating the cooling pump could be an effective strategy to achieve a rapid cold start by significantly reducing thermal mass. Strategies for improving cold start performance at the system level are essential for practical applications in FCVs.

As discussed above, the key to understanding and improving cold starts is through the voltage failure mechanism due to water freeze. Although plenty of output performance experiments have been done in this area, the mechanism of water freezing remains hotly debated. Therefore, it is necessary to conduct further experiments which focus on observations inside electrodes.

2.2. Visualization of liquid and ice

Although output voltage, current, and temperature measurements provide information about cold start performance, these alone cannot present the detailed phenomena occurring inside the cell. Therefore, to better understand the mechanism of performance deterioration in cold starts, it is necessary to analyze ice formation [6,28–30,32,33,52–54]. Diagnostic techniques, such as X-ray imaging, neutron radiography, optical visualization, and magnetic resonance imaging, have been widely used to observe water transport in PEM fuel cells [55]; however, better visualization of ice formation still requires technique improvements. Liquid and ice visualization techniques can be categorized into three levels, based on scope. The first level is micrography with transparent cells [32,52]. The second is indirect imaging with neutron radiography [28–30,56], or X-ray tomographic microscopy [57,58]. The third is scanning electron microscopy with a cryogenic stage and sample preparation unit (CRYO-SEM) [34,47,59–66].

A transparent single cell consists of all the essential components of a regular single cell. Specially, a photographing window is constructed from the outside to the MEA. Ishikawa et al. [6] used 18 μm thick Gore-Select membrane and GDLs based on carbon paper that were PTFE-proofed and therefore hydrophobic. To directly visualize ice formation on the CL surface, the GDL is made with silver mesh rather than carbon paper to allow direct observation of the CL surface through the open area of silver mesh. The flow channel is machined on the stainless steel current collector and is covered by a transparent polycarbonate plate. These components are all sandwiched between the endplates with an open window. This study showed that given certain conditions, liquid water can stay supercooled and can flow out of the CL and GDL. When supercooled droplets grow larger than a certain critical value, they suddenly freeze. Similar results are found in [21], which showed that ice stays within the CL at lower temperatures (-20°C), while ice is more likely to present in both the CL and GDL at a higher temperature (-10°C).

A problem with the above visualization technique is that using silver mesh as GDL does not accurately reflect the realistic condition of a cell due to increased contact resistance. Fuel cells with silver mesh GDLs exhibit much lower polarization performance during startup. Additionally, water access is increased with silver mesh, enough that the possibility of water flooding is lowered compared to a carbon GDL. This shortcoming is avoided in the study by Ge and Wang [32], where carbon GDL is used rather than silver mesh.

Optical access is provided by punching 400 μm holes in the GDL. Observations from this study supported the view that ice forms without supercooled water. They found only liquid water at a relatively high temperature of -3°C . In the initial stage (within 90 s), when the produced amount of water is small, no liquid was found on the surface of the CL. Liquid droplets then emerged on the CL surface, and their size grew with time. When the start temperature was lower than -3°C , only ice was to be found on the CL surface at any time during the process.

The contradictory observations in studies [6,32] can be explained by the fact that supercooled water is very sensitive to peripheral conditions. The presence of supercooled liquid water depends on many factors, such as substrate surface hydrophobicity and nuclei and environmental conditions [67]. Theoretically, if no nucleus were to exist, the liquid phase would remain until the temperature drops to the crystal homogeneous nucleation temperature, which is -48.3°C at standard pressure [68]. Conditions are varied in different experiments or cases with regards to the fuel cell cold start. Even though macroscopic conditions, such as initial temperature, operation current density, and component properties, can be the same among different experiments, it is still impossible to guarantee that all conditions will remain the same at the microscopic level. For example, other important factors include fragments in catalyst layers, size distributions of porous materials, and the hydrophobicity of material surfaces that are in contact with produced water. These can act as the trigger for supercooled water. The cold start is a very complex process in which specific conditions are difficult to control. Therefore, it stands that the presence of supercooled water is a random phenomenon in fuel cell cold starts. As a result, not all the cold start experiments observed the existence of supercooled water. Details of the supercooling phenomenon are further discussed in Section 4.

Although the micrography and transparent cell techniques provide direct information about ice formation, they do have their disadvantages. Because water states cannot be recorded continuously, the phase transition point cannot be measured. Therefore, other techniques, like neutron imaging [54] and X-ray tomographic microscopy [57,58], are needed to measure the liquid-to-ice transition. The neutron imaging technique has several advantages. First, it is non-destructive. Common fuel cell materials are highly transparent to neutrons, but neutrons are heavily scattered by hydrogen-containing compounds. Therefore, water in PEM fuel cells can be easily detected [54,69]. This technique was first used in PEM fuel cells in 1999 [70]. It is additionally capable of detecting the phase change from liquid to ice. Thus, neutron imaging is also suitable for applications in studying cold starts [28–30,54,56].

A dual spectrum neutron radiography technique was developed to directly measure the existence of supercooled water and the transition process from supercooled water to ice [54]. Liquid and ice can be visualized in the in-plane cross section [28,29] or through-plane cross section [30]. Observation of the in-plane cross-section showed that there was more ice under the channel region than in the region under ribs [28]. Oberholzer et al. [29] proved the presence of supercooled water during cold starts by observing the liquid water not only in the GDL but also in flow channels at subfreezing temperatures. They also reasoned that voltage failure is caused by the sudden freeze of supercooled water. Through-plane detecting found that the peak of ice thickness exists on the cathode side of the MEA [30]. Researchers found a non-uniformity of ice distribution in the CL and reasoned that this may be due to the non-uniform reaction rate along the through-plane direction. Thus, water production rate was non-uniform. They also compared dry and wet initial conditions and different operation current densities and found that the MEA could hold more ice before shutdown occurred, with a lower initial water content; less water produced a lower current density.

Although neutron imaging is convenient for water visualization and phase transition detection, the resolution is too low to measure ice distribution inside the CL. To observe more details inside the CL, CRYO-SEM was used. Four SEM images were taken during the cold start process at a constant current [61]. It was found that the ice growth process differs by the magnitude of current density, although the final ice formations were similar. At a higher current density (0.04 A/cm^2), ice initially formed near the membrane side and propagated towards the GDL side to fully cover the CL with ice. At a lower current density (0.01 A/cm^2), ice grows uniformly during the cold start, and the final amount of ice is larger than that of the high current density.

In the same vein, cold start experiments were conducted with constant voltage [34]. The ice growth processes differed from those reported by the study mentioned above [61]. At the beginning of the cold start, ice gradually accumulated in the CL. It was observed that the porosity of the CL gradually decreases. Ice grows faster on the side closer to the GDL, while there is less ice on the membrane side. This ice distribution is contrary to the case with higher current density in [61]. This experiment was conducted at a constant voltage (0.12 V) and current density decreased gradually during the cold start. It follows that the heat generation rate is higher in the initial stage, and there is more heat generation near the membrane. Thus, the membrane side has a higher temperature in the prior stage. This phenomenon possibly suppresses ice formation in the region near the membrane.

2.3. Freeze damage and degradation

Cold starts might result in the degradation of fuel cells. The main cause of degradation is damage to the key components, such as the membrane and catalyst layer, due to water freezing. Alink et al. [35] studied the degradation aspects of a 6-cell stack with -40°C freeze/thaw cycles and cold start. Freeze/thaw cycles with dry and wet conditions were compared. The dry state cold start results in little degradation, while the wet state cold start causes severe performance degradation. By measuring the high-frequency resistance of the stack, it is concluded that the freeze/thaw cycles result in increased flooding; after wet cycles, the stack's performance greatly deteriorates.

Membrane degradation is a vital issue for PEM fuel cells [71,72]. The freezing-induced thermal degradation of the membrane commonly occurs on the membrane surface, which is the interface to the catalyst layer. Since most of the water inside the membrane is trapped in the polymer chains, its freezing point is substantially lowered. Hence the subfreezing operation does not notably change this part of water in the membrane. Water freezing is more likely to occur on the membrane surface, when the bulk water inside the membrane flows out [35]. Ice formation on the membrane surface leads to a variety of problems: expansion of the space between the membrane and catalyst layer, an increase in electric resistance, a slowing down of water uptake from the catalyst layer to membrane, and more seriously, peeling of the catalyst layer and damage of the membrane surface [16,73]. The mechanical property of membrane can also be changed because of the cold start [74]. The membrane may become more fragile under the clamping force.

As for the catalyst layer, the heating/cooling rate is one of the key factors that determines degradation [75]. SEM imaging shows that catalyst layer cracking is mitigated by an optimized purge process before cold start cycles. However, there are still plenty of platinum particles migrating from the CL into the membrane. Phase change in the CL influences cold start ability and also contributes to CL degradation [38].

For the GDL, ice formation in the pores may result in the change of its porous structure. It can change the properties of the GDL, such as the permeability of gas [40] and hydrophobicity [41]. These

properties are essential for the performance of water draining and transport of reactants. In [41], it is found that the contact angle of GDL changes from 131° to 112° after 10 cold start cycles from -10°C . In turn, the hydrophobicity also has a strong effect on cold start performance [76].

2.4. Cold start mechanisms

This subsection will systematically explain cold start mechanisms revealed through experimentation. Typically, initial conditions of the cold start entail a dry membrane and the lack of ice. Because a purge must be done upon shutdown of the fuel cell, residual water content is low. When the cold start begins, dry gases are supplied to flow channels. Then, current builds up as electrochemical reactions occur in the catalyst layer. Water in the membrane tends to move from anode to cathode due to the EOD effect. As a result of the oxygen reduction reaction, water is produced in the cathode. The cathode side consequently accumulates water as a result of both the EOD effect and water produced from the reaction. Although the ionomer in the membrane and catalyst layer can uptake water, there is a limited capacity. The ionomer quickly reaches saturation, and the excess water turns to ice. In most circumstances, ice accumulates in the cathode catalyst layer. Some situations observe the presence of ice in the gas diffusion layer and flow channel. Either way, the ice can block the pores through which reactants flow; it may also cover the three-phase points where electrochemical reactions occur. As a result, the cold start fails.

On the other hand, fuel cell operation may also cause heat generation. Cell temperature commonly increases during cold starts, as observed in experiments. The progression of fuel cell temperature is generally a result of competition between heat generation and heat dissipation. Heat generation rate depends on the current density; higher current density results in higher ohmic heat and activation heat generation rates. The thermal mass of fuel cells also influences temperature. If the rate of temperature rise is large enough to exceed the melting point of ice before the ice fully stops the electrochemical reactions, the cold start may still be successful.

2.5. Summary

Experimental studies on PEM fuel cell cold starts can be grouped into three categories: output performance testing, water/ice visualization, and freezing damage analysis. Output performance testing analyzes the effects of various design parameters and conditions on the performance of cold starts. The evolution of voltage and current density is the result of the combined effects of hydration state (resistance), temperature, and ice blockage. Different failure paths were found. At a lower initial temperature, the performance drop in the final stage tends to be slight, whereas at a higher temperature, performance suddenly drops after a period of steady operation. Non-uniform ice distribution in the catalyst layer and the supercooling phenomenon can also lead to rapid failure.

Second, water/ice visualization experiments seek to uncover the mechanism of performance deterioration due to ice formation. Based on existing studies in the literature, the ice formation process during the cold start can be summarized as follows. The water produced in the CL is initially absorbed by the ionomer. Ice then starts to emerge upon water saturation. In high-current conditions, more water is driven to the membrane's cathode side, so ice emerges first on the near-membrane side of the CL. With a low current density, the membrane tends to take water from the CL, rather than driving water into it. In this case, ice emerges first on the near-GDL side of the CL. From an in-plane view of the section, ice emerges at the under-channel region early. As the fraction of ice increases, pores in the CL are gradually blocked until the cold start fails. Theoretically, if no nucleus existed, supercooled water could sustain a liquid state until the

temperature dropped to the crystal homogeneous nucleation temperature (-48.3°C) [68]. During the cold start, liquid water in a supercooled state flows to the GDL and droplets grow. Once droplet size reaches a critical value, it immediately freezes. As a result, the voltage falls suddenly after a period of steady operation.

Third, the effects of cold starts on fuel cell durability can be elucidated by analyzing the freeze damage to components. Ice formation causes degradation to the membrane interface and catalyst layer due to the inherent volume expansion of water. An effective purge process before the cold start could mitigate damage to the catalyst layer.

3. Overview of models

Modeling the PEM fuel cell cold start process necessitates a combination of the transport phenomena in all components. Analytical and numerical methods in cold start modeling have their respective advantages. The analytical model efficiently reveals the key factors

influencing cold-start performance, while numerical models can quantify details. Cold start simulation models are summarized in Table S2 [5,77–98] in the Supplementary Materials. In this table, models are categorized by objects, dimensions, and methods. Each classification scheme approaches a unique angle, and a comprehensive discussion will follow descriptions of the analytical and numerical methods.

3.1. Analytical model

Analytical models of single-cell cold starts can be found in [77–79]. Fig. 6 summarizes the three major elements of an analytical model: water balance, energy balance, and performance prediction. Key assumptions and equations are presented accordingly. The analytical method for fuel cell modeling is essential for building fundamental understanding of the underlying processes by revealing parametric dependencies [99]. Therefore, it provides a general understanding of fuel cell operation and design parameters.

	Schematic Diagram	Key Assumptions	Key Equations
Water Balance		- Lumped system - Ideal gas - Isotropic and homogeneous materials - Instantaneous ice formation from vapor upon saturation - No ice in GDL & flow channel - Outflow saturated with water vapor	- Overall water balance: $\dot{n}_{H_2O}^{acc} = \dot{n}_{CL}^{H_2O} - \dot{n}_{mem}^{H_2O} - \dot{n}_{GDL}^{H_2O}$ - Ice in CL: $\int_0^{t_0} \dot{n}_{acc}^{H_2O} dt = \frac{\rho_{dry}(\lambda_{sat} - \lambda_0)}{EW} \epsilon_e \delta_{CL} A$ - Ice volume fraction: $s = s_0 + \int_{t_0}^t \frac{\dot{n}_{acc}^{H_2O} v_{ice}}{\epsilon_{CL} \delta_{CL} A} dt = s_0 + \int_{t_0}^t \frac{(\dot{n}_{CL}^{H_2O} - \dot{n}_{mem}^{H_2O} - \dot{n}_{GDL}^{H_2O}) v_{ice}}{\epsilon_{CL} \delta_{CL} A} dt$
Energy Balance		- Uniform temperature - Negligible heat of water absorption in the membrane	- Overall heat balance: $m_{thermo} \cdot \frac{dT}{dt} = Q_{gen} + Q_{loss}$ - Total heat generation: $\dot{Q}_{gen} = \left(U_o + \frac{h_{ph}}{2F} - T \frac{\partial U_o}{\partial T} \right) IA - U_{cell} IA$ - Total heat loss: $\dot{Q}_{loss} = (\dot{m} C_p)_{outlet} T - (\dot{m} C_p)_{inlet} T_0 + h_0 A (T - T_0)$
Performance Prediction		- Ice hinders oxygen transport and reduces the active catalytic site, thus leads to voltage loss - Linear relationship is assumed between the fraction of surface coverage and the ice volume fraction - Negligible voltage loss due to electronic resistances in CL, GDL, and BP	- Output voltage: $V_{out} = V_{nernst} + V_{act} + V_{conc} + V_{ohmic}$ - Open circuit voltage: $V_{nernst} = 1.23 - 0.9 \times 10^{-3} (T - T_0) + \frac{RT_0}{2F} \ln \left(\frac{p_{H_2} p_{O_2}}{p_{H_2O}} \right)$ - Activational voltage loss: $V_{act} = -b \ln \left(\frac{I}{(1 - s_{ice} - s_{lq})^{0.5} j_*^h} \right)$ - Concentration voltage loss: $V_{conc} = b \ln \left(1 - \frac{I}{j_D} \right)$ - Ohmic voltage loss: $V_{ohmic} = - \left[(ASR_{BP} + ASR_{GDL} + ASR_{mem}) I + \frac{ASR_{CL} I}{3} \right]$

Fig. 6. Schematic diagram, key assumptions and key equations of analytical models for PEMFC cold start.

The single-cell cold start model is capable of describing the water, heat, and output performance, which are three major problems that arise during the cold start process. As shown in Fig. 6, most analytical models for the single-cell cold start focus on inter-layer water exchanges. Adversely, the existence of supercooled water is commonly neglected, along with ice formation in the GDL and flow channel, since a negligible amount of water can flow out of the CL without the presence of supercooled liquid. Some theoretical models focus on supercooling phenomenon [98] and ice formation in the GDL [100], but since these models are for specific transport phenomena rather than for a cold start model, they are discussed in detail in Section 4.

A water balance model is constructed, as illustrated in the first schematic diagram in Fig. 6. The water produced has two paths to move out of the cathode CL; it can diffuse to the membrane through ionomer, or it evaporates and diffuses to the GDL and flow channel until it is carried away by the gas flow. The excess water accumulates in the cathode CL as ice. Therefore, the formulae of water balance are given as follows:

$$\dot{n}_{acc}^{H_2O} = \dot{n}_{CL}^{H_2O} - \dot{n}_{mem}^{H_2O} - \dot{n}_{GDL}^{H_2O} \quad (1)$$

where the three terms on the right-hand side are given as:

$$\dot{n}_{CL}^{H_2O} = \frac{IA}{2F} (1 + n_d) \quad (2)$$

$$n_{mem}^{H_2O} = \frac{\rho_{dry}(\lambda_{sat} - \lambda_0)}{EW} \delta_{mem} A \quad (3)$$

$$\dot{n}_{GDL}^{H_2O} = \dot{n}_{outflow}^{H_2O} = \zeta_c \frac{IA}{4F} \frac{1}{0.21} \frac{p_{sat}^{H_2O}}{p_c} \quad (4)$$

Ice in the cathode CL appears after the CL ionomer is saturated. By defining the saturated moment as t_0 , the following equation can be obtained:

$$\int_0^{t_0} \dot{n}_{acc}^{H_2O} dt = \frac{\rho_{dry}(\lambda_{sat} - \lambda_0)}{EW} \varepsilon_e \delta_{CL} A \quad (5)$$

The ice volume fraction is calculated as:

$$s = s_0 + \int_{t_0}^t \frac{\dot{n}_{acc}^{H_2O} v_{ice}}{\varepsilon_{CL} \delta_{CL} A} dt = s_0 + \int_{t_0}^t \frac{(\dot{n}_{CL}^{H_2O} - \dot{n}_{mem}^{H_2O} - \dot{n}_{GDL}^{H_2O}) v_{ice}}{\varepsilon_{CL} \delta_{CL} A} dt \quad (6)$$

In an analytical model, the simplest treatment of energy balance is to obtain a lumped thermal model, as illustrated in Fig. 6. It assumes a spatial uniformity of temperature through the cell. The general energy balance can be presented as:

$$m_{therm} \cdot \frac{dT}{dt} = Q_{gen} + Q_{loss} \quad (7)$$

where the total heat generation is

$$\dot{Q}_{gen} = \dot{Q}_{rev} + \dot{Q}_{irrev} + \dot{Q}_{pc} \quad (8)$$

The total heat generation consists of three heat sources: reversible heat, irreversible heat, and heat release due to phase change.

$$\dot{Q}_{rev} = (-T\Delta S) \frac{I}{2F} A = \left(-T \frac{\partial U_0}{\partial T}\right) IA \quad (9)$$

$$\dot{Q}_{irrev} = \left(-\frac{\Delta G}{2F} - U_{cell}\right) IA = (U_0 - U_{cell}) IA \quad (10)$$

$$\dot{Q}_{pc} = \dot{n}_{ice}^{H_2O} \cdot h_{pc} \quad (11)$$

By combining the above equations, the total heat generation can be written as:

$$\dot{Q}_{gen} = \left(U_0 + \frac{h_{pc}}{2F} - T \frac{\partial U_0}{\partial T}\right) IA - U_{cell} IA \quad (12)$$

The heat loss is mainly caused by enthalpy differences between the inlet and outlet gases and heat dissipation through various surfaces. Hence, the heat loss equation is written as:

$$\dot{Q}_{loss} = (\dot{m}C_p)_{outlet} T - (\dot{m}C_p)_{inlet} T_0 + h_0 A (T - T_0) \quad (13)$$

Thirdly, common indicators for performance prediction are voltage and current density output of the single cell. Fig. 6 shows that performance may degrade in different patterns during the cold start due to differences in hydration conditions, temperatures, and ice blockages of the membrane. It is important to discern how these operation parameters affect performance.

In an analytical model, charge conservation equations are usually neglected. While the Tafel representation performance formulation is used as an alternative, the output voltage can be expressed as:

$$V_{out} = V_{nernst} + V_{act} + V_{conc} + V_{ohmic} \quad (14)$$

The four terms on the right hand side are the open circuit voltage (OCV), activation voltage loss, voltage loss due to reactant transport, and ohmic voltage loss.

$$V_{nernst} = 1.23 - 0.9 \times 10^{-3} (T - T_0) + \frac{RT_0}{2F} \ln \left(\frac{p_{H_2} p_{O_2}}{p_{H_2O}} \right) \quad (15)$$

$$V_{act} = -b \ln \left(\frac{I}{(1 - s_{ice} - s_{lq})^{0.5} j_*^h} \right) \quad (16)$$

$$b = \frac{RT}{\alpha F} \quad (17)$$

$$j_*^h = j_* \delta_{CL} \left(\frac{C_h}{C_{ref}} \right) \quad (18)$$

$$c_h = \frac{1 + (1 - 1/\zeta) 0.21 p_c}{2 RT} \quad (19)$$

The effects of ice and liquid accumulation on the activation voltage loss are considered by using the ice and liquid volume fractions:

$$V_{conc} = \frac{RT}{\alpha F} \cdot \ln \left(1 - \frac{I}{j_D} \right) \quad (20)$$

$$j_D = \frac{4Fc_h}{\delta_{GDL}/D_{GDL}^{eff} + 0.5\delta_{CL}/D_{CL}^{eff}} \quad (21)$$

$$V_{ohmic} = - \left[(ASR_{BP} + ASR_{GDL} + ASR_{mem}) I + \frac{ASR_{CL} I}{3} \right] \quad (22)$$

The area specific resistance (ASR) is calculated as

$$ASR = A_{cell} \cdot R \quad (23)$$

This model can predict the effects of ice volume fraction on the cathode overpotential [94]. The overpotential decreases linearly with ice volume fraction, as voltage loss is primarily due to the activation loss caused by the blockage of reaction sites at that stage. When the ice volume fraction approaches unity, air transport pores in the cathode CL rapidly close, causing the resistance of mass transport to sharply increase and resulting in the sudden drop of overpotential.

Results of analytical models reveal cell voltage, temperature, and ice volume fraction performance as functions of time. Typically, in a failed cold start, voltage first rises due to the hydration of the membrane. During the last several seconds, cell voltage suddenly drops when the ice volume fraction is close to unity. Ice is formed after the ionomer of the cathode CL is saturated with non-frozen membrane water. The temperature of the fuel cell begins to rise after a cold start. Analytical results can also clearly show the relationship among parameters, providing a way to identify parameters that affect cold-start performance. For example, a key study found that the water

removal/accumulation ratio (β) is the most critical parameter for cold-start performance; if the value of β is less than unity, it is indicative of water accumulation in the cathode CL [77]. Otherwise, water is drained. β is influenced by operation conditions, such as temperature, stoichiometric ratio, and pressure. The gas flow's ability to remove water via evaporation is weak, and it requires a relatively high temperature and high flow rate to make β reach unity.

3.2. Numerical model

Fig. 7 shows a general description of numerical models for the PEM fuel cell cold start. Conservation equations for gas mixtures in the anode and cathode are solved and include conservation of mass, momentum, and gas-phase species, with the assumption that the gas is an ideal gas and the flow is laminar. Several assumptions may

	Schematic Diagram	Key Assumptions	Key Equations
Mass Conservation		- Solved in flow channel, CL and GDL - Incompressible and ideal gas - Isotropic material properties	$\frac{\partial}{\partial t}(\varepsilon(1-S_{lq}-S_{ice})\rho_g) + \nabla \cdot (\rho_g \vec{u}) = S_m$
Momentum Conservation		- Solved in flow channel, CL and GDL - Incompressible and ideal gas - Isotropic material properties - Laminar flow	$\frac{\partial}{\partial t} \left(\frac{\rho_g \vec{u}_g}{\varepsilon(1-S_{lq}-S_{ice})} \right) + \nabla \cdot \left(\frac{\rho_g \vec{u}_g \vec{u}_g}{\varepsilon^2(1-S_{lq}-S_{ice})^2} \right) = -\nabla P_g + \mu_g \nabla \cdot \left(\nabla \left(\frac{\vec{u}_g}{\varepsilon(1-S_{lq}-S_{ice})} \right) + \nabla \left(\frac{\vec{u}_g^T}{\varepsilon(1-S_{lq}-S_{ice})} \right) \right) - \frac{2}{3} \mu_g \nabla \cdot \left(\nabla \cdot \left(\frac{\vec{u}_g}{\varepsilon(1-S_{lq}-S_{ice})} \right) \right) + S_u$
Gas Species Conservation		- Solved in flow channel, CL and GDL - Incompressible and ideal gas - Isotropic material properties - Laminar flow	$\frac{\partial}{\partial t}(\varepsilon(1-S_{lq}-S_{ice})\rho_g Y_i) + \nabla \cdot (\rho_g \vec{u}_g Y_i) = \nabla \cdot (\rho_g D_i^{eff} \nabla Y_i) + S_i$
Water Conservation	Vapor to ice assumption	- Instantaneous desublimation of water from supersaturated vapor - No liquid exists	$S_c^{H_2O} = -\frac{S^{H_2O} j}{nF} - \nabla \cdot \left(\frac{n_d}{F} I_e \right) + q_{sg}^{H_2O}$
	Membrane water to ice assumption	- Membrane water transfer to ice when gases are fully humidified - Membrane water transfer to vapor with non-humidified gases - Vapor transfer to liquid or ice upon saturation - Liquid water freeze to ice given subfreezing temperature	$\frac{\partial(\varepsilon S_{lq} \rho_{lq})}{\partial t} + \nabla \cdot (\rho_{lq} \vec{u}_g) = \nabla \cdot (\rho_{lq} D_{lq} \nabla S_{lq}) + S_{lq}$ $\frac{\partial(\varepsilon S_{ice} \rho_{ice})}{\partial t} = S_{ice}$ $\frac{\rho_{mem}}{EW} \frac{\partial(\omega \lambda_{rf})}{\partial t} = \frac{\rho_{mem}}{EW} \nabla \cdot (\omega^{1.5} D_{nmv} \nabla \lambda_{rf}) + S_{nmv}$
	Supercooled water to ice assumption	Supercooled liquid water initially forms and transfer to ice when exceeds critical cluster radius	$r^* = -2\sigma_{cl} \frac{T_e}{\rho_c \cdot L \cdot \Delta T}$
Charge Conservation		- Frozen membrane water does not contribute to the ion transport - The effective electron and ion conductivities are calculated based on the Bruggemann correlation	$0 = \nabla \cdot (\kappa_{ele}^{eff} \nabla \phi_{ele}) + S_{ele}$ $0 = \nabla \cdot (\kappa_{ion}^{eff} \nabla \phi_{ion}) + S_{ion}$
Energy Conservation		The different phases are at local thermodynamic equilibrium, therefore a single energy conservation equation is solved	$\frac{\partial}{\partial t}((\rho C_p)^{eff}_{fl,sl}) + \nabla \cdot ((\rho C_p)^{eff}_{fl} \vec{u}_g T) = \nabla \cdot (\kappa_{fl,sl}^{eff} \nabla T) + S_T$

Fig. 7. Schematic diagram, key assumptions and key equations of numerical models for PEMFC cold start.

be made about water conservation. The first one assumes instantaneous desublimation of water from supersaturated vapor. The second assumes membrane water changes to ice or vapor depending on gas humidity. The third assumes that the supercooled water is an intermediate between membrane water and ice. The first two assumptions are commonly used in existing cold-start models. Supercooled water has been observed in recent experimental studies, but mechanisms remain debated. Details of the supercooling phenomenon are discussed in Section 4. Charge conservation is considered for all fuel cell components. It is assumed that frozen water in the membrane does not contribute to ion transport. Energy conservation deals with heat conduction and convection in different components. It is assumed that phases are at local thermodynamic equilibrium.

Numerical procedures for cold-start models generally follow a few typical approaches for PEM fuel cell modeling, as summarized in [101]. They are commonly implemented in commercial computational fluid dynamics (CFD) software with customized development, such as the user-defined function (UDF) for FLUENT [5,80,85,88], or COMSOL [82]. The discretization scheme is usually based on the finite volume method (FVM), as well as the finite element method (FEM) [82]. After discretization, a Gauss-Seidel iterative solver can be used to solve the complex linear systems. Since cold-start models are usually constructed in a transient state, the PISO algorithm is commonly used to solve the Navier–Stokes equations, as well as the SIMPLE algorithm [84]. A special issue for cold-start modeling is the treatment of phase change in various areas, such as the presence of ice in the CL/GDL and the freezing of membrane water in ionomer. These equations are usually coded through user-defined interfaces.

3.2.1. One-dimensional numerical model

One-dimensional models are commonly used for simulating cold starts at the stack level. Since temperature affects the cold start in a PEM fuel cell stack, the core idea of this model is to predict the stack's cold-start performance based on temperature distribution. Lower computational costs present a major advantage of these simplified models; additionally, simulations can be done on small stacks or practical, large-scale stacks with hundreds of cells, while making minute variations to the model [92,93].

An early thermal model [92] adopted the layered modeling approach. This model could be considered a quasi-one dimensional model. Each cell in the stack is separated into layers, including the coolant, plate, anode gases, anode GDL, anode CL, membrane, cathode CL, cathode GDL, cathode gases, and cathode plate. At the two ends of the stack are the endplates. The entire stack is constructed by repeating single-cell layers. Temperature distribution can be obtained from this model, without having to solve for differential equations. Khandelwal et al. [93] developed a one-dimensional model that included conduction and convection heat transfer. In this model, start-up time is defined as the time when temperatures in all regions exceed 0 °C. Based on this indicator, various conditions are simulated, such as operation current density, inlet gas temperature, and external heating. Start-up time and energy consumption with respect to the number of stack cells are analyzed using this model. It was found that start-up time and normalized energy decreased sharply as the number of cells in a small stack increased (fewer than 10 cells). In stacks containing more than 20 cells, the number of cells no longer had a notable influence on cold-start characteristics.

A one-dimensional stack model was developed that accounts for multi-phase transport phenomena, including heat transfer, membrane water transport, liquid water transport, ice formation, and gas diffusion [94]. This model provides more precise predictions of cold-start performance and more details about transport phenomenon, compared to early thermal models. Furthermore, since all governing equations are solved in the through-plane direction, it also has high computational efficiency. It therefore is easily applied to large-scale

stacks. Some relevant simulation results based on this model are discussed as follows.

The effects of stack size on the start duration and maximum stack temperatures in failed cold start processes are shown in Fig. 8(a). Start duration is defined as the time from the beginning of a cold start to the shutdown moment; all test cases led to failure, as seen in Fig. 8(b). The start duration also describes the time when the stack reaches its maximum temperature, since temperature stops increasing upon shutdown. The temperatures shown in this figure illustrate

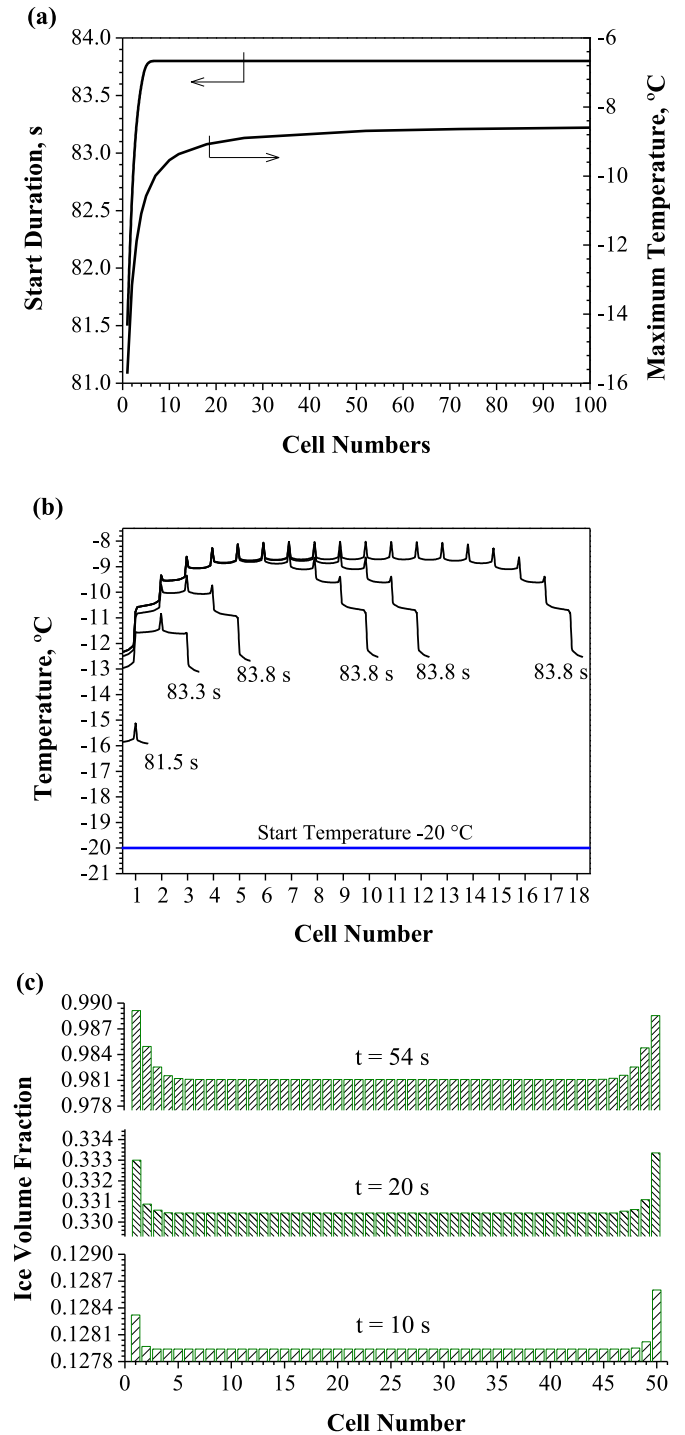


Fig. 8. Cold start characteristics of PEMFC stacks. (a): Temperature distribution along the through-plane direction; (b): Cold start duration and maximum stack temperature as a function of number of cells in a stack; (c): Evolution of ice distribution in a 50-cell stack (adapted from [94]).

maximum temperatures that correspond to stacks with various amounts of cells. Stacks with more cells appear to sustain the cold-start process for longer and can reach a higher final temperature. In terms of stack size, the heat generation rate is proportional to the volume of the stack, while the heat dissipation rate is proportional to the surface area of the stack. Larger stacks have smaller specific surface area and therefore smaller heat losses. As a result, larger stacks tend to be better insulated. This effect is more significant when the cell number is under five. It is observed that compared to a single cell, a 5-cell stack can increase the cold-start duration by 2 s and obtain a 5 °C higher final temperature. However, in stacks with more than 20 cells, both the cold-start duration and final temperature tend to be steady, even as cell number increases. This observation is consistent with the results in [93]. Temperature distribution along the through-plane direction, among stacks of different cell numbers, are shown in Fig. 8(b). Generally, cells in the middle region have higher temperatures due to heat dissipation occurring at the endplates. The temperature differences between the peak value in MEA and valley value in flow channels vary based on conditions and material properties. Fig. 8(c) shows the evolution of ice volume fraction distributions in the 50-cell stack. The cold-start is at a constant current of 0.15 A cm^{-2} . More ice accumulated in the side cells due to the lower temperature in these regions. The variation in the amount of ice from the side cells to the middle cells tended to increase over time due to an increase in the temperature gradient.

3.2.2. Multi-dimensional multiphase model

Multiphase cold-start models in two dimensions [80–83,102] or three dimensions [5,84–89,103,104] allow for more precision during simulation. They reveal more details about transport phenomena than one-dimensional models, especially for anisotropic materials. The major features of multi-dimensional, multiphase models are that they are capable of resolving the effects of spatial distribution of water, ice, temperature, and their couplings on cold-start performance. The transient temperature distribution in the through-plane cross-section of a straight channel single cell is shown in Fig. 9 [5].

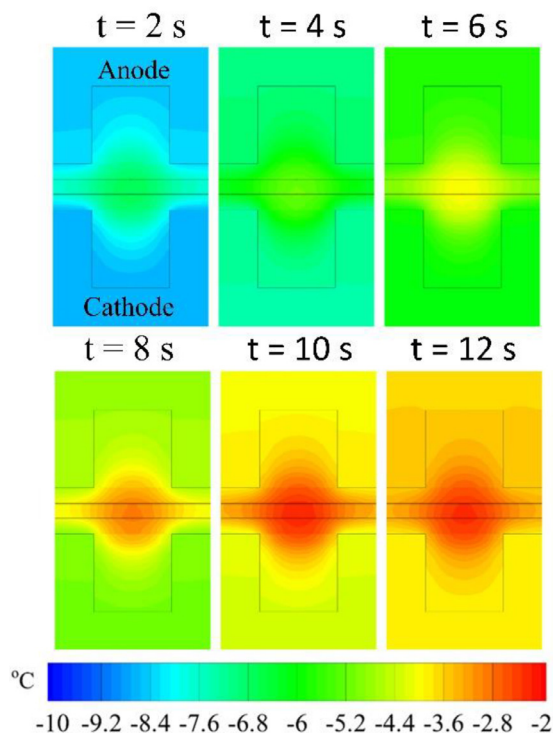


Fig. 9. Transient temperature distribution in a through-plane cross section of a straight-channel PEMFC (adapted from [5]).

The highest temperature was located in the MEA under-channel region, from being directly heated up by ohmic heat produced in the dry electrolyte and the reaction heat generated in the CLs. The temperature in the anode side was higher than the cathode side because the former's catalyst layer was dryer than the latter's. In addition, due to the higher thermal conductivity of hydrogen compared to oxygen, the heat transfer rate is higher in the anode.

The temperature distribution is also affected by the rate of heat generation from various sources. Typically, the largest source of heating is ohmic heat, resulting from the low electric conductivity and the relatively high current density of the membrane during constant voltage operation. If the current density is low, ohmic heating would be insignificant. In comparison, activation heating can be higher than ohmic heating [84]. Reversible heat and latent heat are much lower than activation heat. After the cathode CL is fully obstructed by ice, all the heating sources disappear as reactions terminate and current density drops to zero. Immediately after that point, cell temperature reaches its peak point.

The evolution of water in the cathode CL, anode CL, and membrane is different due to the combined effects of EOD and back diffusion. Before the cathode CL ionomer is saturated, water content in the cathode CL sharply increases. In the same stage, membrane water in the cathode CL is not only supplied by the EOD effect, which drives water from the anode to cathode, but also by the water reactant generated in the cathode CL. The decrease of water in the anode CL is a result of competition between the EOD effect and back diffusion effect. After the cathode CL ionomer is saturated, the rate of water increase slows down in the cathode CL.

Fig. 10 shows the transient ice distribution in the cathode CL in the through-plane direction. Ice first formed in the middle of the CL. In the region close to the membrane, ice does not easily form, since water in this region tends to diffuse into the membrane due to a large gradient. In the region near the flow channel, water can more easily be taken by the dry air flow. As a result, the rate of ice formation is lower in this region. The temperature in the under-land region is lower than that of the under-channel region because the former is closer to the surrounding walls. As a result, it was observed that ice first emerges under the land and then propagates to the under-land region. Eventually, the cathode CL is fully covered by ice, indicating the failure of the cold start process.

As the stack level, multi-dimensional models are rare in the literature due to the necessity of heavy calculations. A two-dimensional PEM fuel cell stack model was developed in [95]. This model considers the formation and melting of ice, species transport, and current distribution. The implicit finite-difference method was used to solve the system of equations. To validate this model, single-cell simulation results are compared with the experiment's results in [105]. The experiment was conducted with a single cell, which had a reaction area of 104 cm^2 and a membrane of $30 \text{ }\mu\text{m}$ Gore-Tex. This model was used to simulate a large-scale stack consisting of 230 cells of 430 cm^2 cross-section area. It was found that a high current density favors a rapid cold start, because more heat can be produced. Thin metal plates, which have lower thermal mass, are also beneficial for a rapid cold start. A three-dimensional stack cold start model was developed in [90]. Cold-start characteristics of a 3-cell stack, a 2-cell stack, and a single cell were compared. The study found that the temperature of the middle cell was higher than the side cells due to its distance from heat dissipation surfaces. The temperature of the anode side of the stack was higher than that of the cathode side because more ohmic heat was produced on the anode side. For stacks of varying cell numbers, average cell voltages were similar at the beginning of the cold start due to similar initial conditions. Voltage discrepancies between different stacks became more notable as ice gradually accumulated in cathode CLs. The voltage of larger stacks decreased more slowly due to a lower rate of ice formation and a higher rate of increasing temperature.

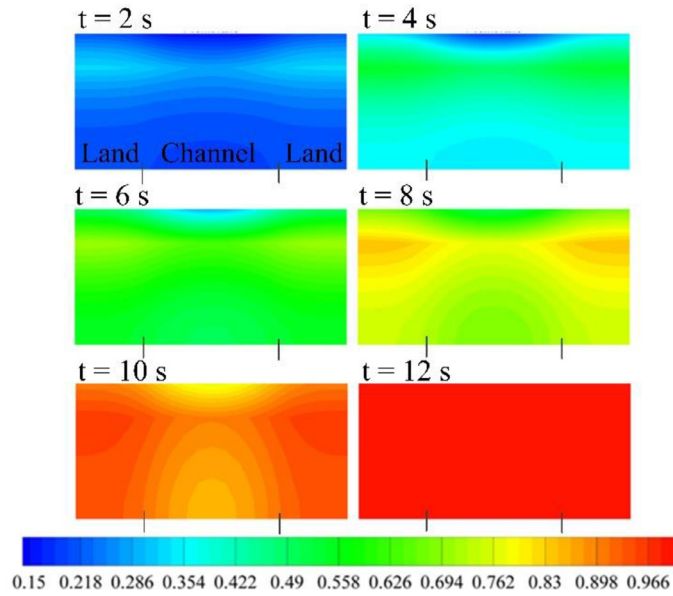


Fig. 10. Transient ice distribution in cathode of catalyst layer in a through-plane cross section at a constant voltage of 0.3 V with Nafion™ 117 (adapted from [5]).

3.3. Microscopic model

Cold-start models for single PEM fuel cells or stacks assume homogeneous cell component materials. However, microscopic models are needed to analyze the effects of component microstructures on freezing and transport processes. To describe the microscopic morphology of the Nafion™ membrane, three types of structural models have been proposed: the cluster-network model [106], rod-like structure model [107], and cylindrical nanochannel model [108]. These models differ in their description of the void patterns of the membrane. The distribution of the hydrophobic backbone of PTFE and hydrophilic sulfonic acid groups is a major factor that determines water transport and phase change features in the membrane at subfreezing temperatures.

The real nanostructure of the membrane, and how it affects water phase changes at subfreezing temperatures, remains debated. The rod-like structure model [107] assumes that the membrane's nanostructure transforms when the membrane switches from the dry state to the hydrated state. Schmit and Chen [108] presented the parallel nanochannel model. The schematic diagram of the nanochannel is shown in Fig. 11. Polymer backbones form nanochannels, which provide pathways for water diffusion. The inner side of the nanochannel is covered with hydrophilic groups. At subfreezing temperatures, water in the larger diameter channel froze, while membrane water in smaller channels remained free. A model based on this theory successfully predicted that Nafion™ maintains conductivity at a very low temperature, which was later supported by experiments.

Three-phase contact interfaces are sites where electrochemical reactions occur and ice forms. Microscopic modeling of ice in the CL can be found in [59,61,109]. A three-phase boundary cathode CL model was developed [59]. Some basic assumptions were made: first, the reaction only occurs with oxygen dissolved in the ionomer. Secondly, mass transport losses are negligible. This model explained the evolution of ice distribution in the CL, as observed in experiments with CRYO-SEM. Two types of physical damage in the MEA due to the cold start were found [110,111]. Diffusion media perforated through and delaminated the catalyst. An analytical model was developed to predict the time of ice melting in the GDL, as a correlation to structural parameters and thermal properties [97].

3.4. Model validation

An important part of modeling the PEM fuel cell cold start is experimental validation. The uncertainties of mathematical models commonly originate from two aspects [112]. One is epistemic uncertainty due to the lack of understanding of certain properties or mechanisms. For example, imperfect modeling methods for heat and mass transfer in porous media can be a source of uncertainty. The epistemic uncertainties also derive from the less precise measurements of material properties, such as conductivity, porosity, permeability, and catalyst reactivity. Environmental variations, as in temperature, are also sources of epistemic uncertainties. The second category is aleatory uncertainty, which stems from the statistical nature of physical phenomena. For the cold start, the stochastic behavior of supercooled water is a major source of uncertainty.

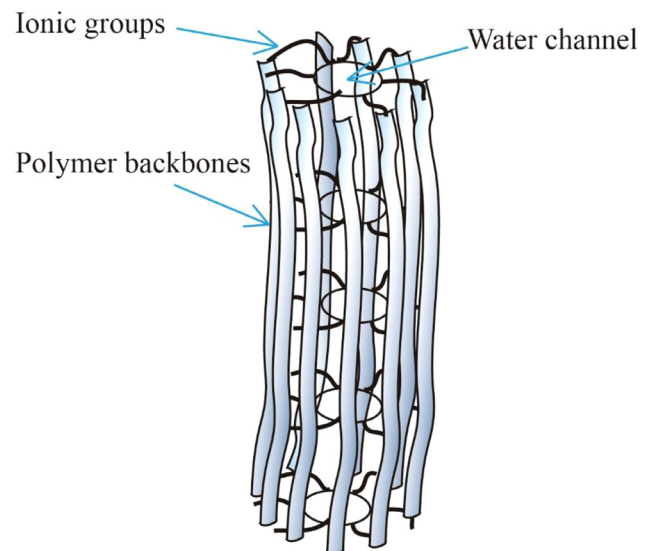


Fig. 11. Schematic diagram of the parallel nanochannel model (adapted from [108]).

In PEM fuel cell cold start model validation, commonly used variables include voltage output, cumulative water amount, and evolution of cell temperature. All multiphase numerical cold-start models reviewed in this section are validated by quantitative comparisons between model predictions and experimental results under certain conditions. Mao et al. [85] used transient voltage and cumulative product water as parameters to validate the three-dimensional multiphase model. For transient voltage, the minimum relative error is around 3% and the maximum relative error is 14%. This accuracy is based on a constant-current cold start at -20°C and 0.04 A cm^{-2} . By comparing the amount of produced water, the model was validated under a wide range of conditions, with variations in cold-start temperature (-20°C to -30°C), current density (0.04 A cm^{-2} to 0.1 A cm^{-2}), membrane thickness ($30\text{ }\mu\text{m}$ to $60\text{ }\mu\text{m}$) and membrane water content (dry purge and equilibrium purge). The result showed that model accuracy was well within $\pm 20\%$ under many conditions.

Balliet and Newman [102] conducted experiments to validate the two-dimensional multiphase cold start model. Validation factors included cumulative water amount, initial cell potential, cell voltage during the cold start, and the evolution of cell temperature. For the amount of cumulative water, errors were generally within $\pm 20\%$. Cold start temperature ranged from -5°C to -30°C and active cell area ranged from 5 cm^2 to 320 cm^2 . For initial cell potential, the maximum error was around 12%. By validating cell voltage and the evolution of temperature, it was found that error increased with the decrease of cold start temperature. When the cold start temperature was between -10°C and -20°C , the error of voltage and temperature was limited to 10% and 15%, respectively. However, when the cold start temperature was -30°C , the maximum error of voltage could be more than 100%, and the temperature error could be around 40%, rendering these modeling results unreliable. These findings suggest that cold start models have their ability boundaries. Model predictions can match experimental results very well within certain conditions, but errors can be of a great magnitude if models are used beyond their range of applicability.

Ko and Ju [88] validated the multiphase cold start model under various conditions. Current density ranged from 0.04 A cm^{-2} and 0.08 A cm^{-2} , cold start temperature varied from -20°C to -30°C , and water content varied from 2.9 to 6.4. Cell voltage was used for the validation. The study found that the model predictions agreed with experimental results within the above-mentioned conditions. The errors were between 3% and 16%, depending on the stage of voltage evolution. Much like the results from [85], it was found that maximum model errors occur when voltage starts to drop due to ice blockage; the model predicted faster voltage drops than the experiment. Nevertheless, the model predicated characteristics of voltage evolution for a failed cold start process and also captured the effects of key conditions, such as initial membrane water content and initial temperature. A similar phenomenon was found for the validation of a one-dimensional stack model (voltage error: 3% to 8%) [94] and for a three-dimensional multiphase single-cell cold start model (voltage error: 3%–5%) [5]. All of these models compared favorably to experiments, except that voltage decay could not be precisely predicted. This issue is mainly due to the difficulty in understanding the actual impact of ice formation on the reactivity of catalyst layers. To improve the cold-start model regarding the voltage decay profile issue, Mishler et al. [30] used high-resolution neutron radiography to observe the ice formation process, with the goal of modifying the numerical model. They found that the ice coverage factor is not a constant, but is dependent on ice content. After this modification, the model could be well-validated by experimental results, and the voltage decay process was better precisely modeled, with less than a 6% error.

Existing analytical models in the literature do not provide direct validation by comparing model results with experimental

results. There are two reasons for this situation. First, for analytical cold start models, many assumptions and simplifications must be made, rendering this model incapable of reproducing the experiment. For example, analytical models commonly assume that ice only forms in the CL, which is treated with a lumped model. However, experiments commonly find that ice accumulates in the GDL and even the flow channel, and that the thickness of the CL also greatly impacts cold-start performance. Therefore, it is seemingly futile to validate these analytical models with direct experiments. The second reason is that the major goal of analytical cold start models is to clearly show parametric dependencies and to provide a general understanding of fuel cell operation and effects of design parameters in order to suggest parametric optimization strategies. Predicting performance or phenomena is not the aim of analytical cold start models. Rather, the reliability of analytical models is based on well-developed knowledge of fuel cells and its transport phenomena. Some microscopic models are focused on modeling component properties under sub-freezing conditions. For example, the model described in [108] is validated by comparing modeling results to experimental results from a small-angle X-ray. Some other microscopic models focus on the ice formation process in porous media [59,61,97,109]. These theoretical models are difficult to validate in direct experiments. Although it is challenging to quantify the accuracy of these theoretical models, they do provide fundamental insights that benefit further in-depth studies.

Four conclusions can be reached from the above discussion. First, multiphase numerical models were all validated by comparing model predictions to experimental results, and within certain conditions, the relative errors were capped at 3 to 20%. This level of accuracy is acceptable for observing the effects of various conditions on cold-start performance. Secondly, future model validation should focus on the critical point at which voltage sharply decreases. Third, analytical cold start models do not commonly provide direct validation through experiments due to the many assumptions made for analytical models, so experimental conditions cannot be reenacted. These analytical models are useful, as they could reveal parametric dependencies and suggest directions for optimization. Fourth, these cold start models have their boundaries; while model predictions can match experimental results reasonably within certain conditions, the magnitude of errors can be unreasonable if the models attempt to simulate conditions outside of their capability.

3.5. Summary

This section reviewed modeling studies on PEM fuel cell cold starts. Models are categorized into three types according to their methods and scopes: analytical, numerical, and microscopic. Analytical models can directly build on correlations between various parameters. Through this, key factors affect the cold start performance can be discovered. In an analytical cold start model, the basic structure consists of water balance submodel, energy balance submodel, and performance prediction submodel. Numerical models can be implemented in different dimensions. One-dimensional numerical models are suitable for simulating fuel cell stacks, as the characteristics along the through-plane direction dominate the cold start. Studies based on one-dimensional stack models found that larger stacks last longer before a cold start failure. Thermal properties also influence cold-start performance. Lower heat capacity and higher conductivity are favored. Multi-dimensional multiphase models can predict water and ice distribution in different dimensions. Thus, they can provide more details for cold-start performance optimization. Microscopic models give insight into the mechanism of water freezing in the membrane and porous layers. The parallel nanochannel model for NafionTM explains the coexistence of frozen

membrane water and free membrane water under subfreezing conditions. The three-phase boundary model for the CL explains the evolution of ice distribution in the CL, as observed in experiments with CRYO-SEM. Model validation has been made for all the multi-phase numerical models discussed in this review, and the relative errors are between 3% and 20%, which are acceptable for the purposes of analyzing the cold-start process. Models should be used carefully by considering the appropriate scope. Further model validation needs to be made with focus on the critical failure stage, when voltage drops quickly.

4. Transport phenomena in cold start

Since the cold-start process involves a wide temperature range, from subfreezing to normal temperatures, the transport phenomena differ from those in normal operation, especially when the phase change of water is involved. In this section, mechanisms of various transport processes during the cold start in PEM fuel cells are analyzed based on experimental observations and modeling studies, as reviewed in earlier sections.

4.1. Water transport

4.1.1. Membrane processes

Water transport in the ionomer is affected by the temperature and water content during the cold start process. There are two aspects to be discussed. One is the phase change from free membrane water to frozen membrane water. The other is the diffusion process of free membrane water. As illustrated in Fig. 1, at normal temperatures, membrane water diffuses through the SO_3^- bond or in its free state according to the saturation states [113]. At a subfreezing temperature, some of the free water becomes frozen membrane water, which does not contribute to proton conduction. The amount of remaining free membrane water is temperature-dependent [56].

There are three types of non-frozen water with different mobilities, which depend on the bonding strength of water to SO_3^- . The type of membrane water with the most mobility is widely considered to be bulk water. The second strongest is free water, which is loosely bound to SO_3^- . It has a slightly lower freezing point than bulk water because it is confined in membrane pores. The third type of water is more tightly bonded to hydrophilic groups. Hence it can induce a notable freezing point reduction, although it is still freezable. It is possible for all the above water freezes to occur at subfreezing temperature. Another type of water is non-freezable water, which is most tightly bound to SO_3^- . This portion of water does not freeze as temperature decreases, and therefore is always contributing to proton transfer. This type of water has a saturation limit of around 4.8 water content.

When a cold start process initiates, certain regions of the membrane become hydrated due to the EOD effect and water generation. If the initial water content is very low, the water initially accumulated becomes non-freezable water until it reaches the saturation limit. When this moment occurs, the produced water becomes non-frozen membrane water until saturation. The content of saturated water is governed by temperature.

The water diffusion coefficient in membrane depends on the water content in the membrane. The correlation was determined by experiments [114,115]. Pulsed gradient nuclear magnetic resonance (NMR) spectroscopy is usually used to measure the water diffusion coefficient. There are two correlations commonly used for PEM fuel cell cold-start modeling. They are listed as Eqs. (24) and (25) [114,115].

$$D_{nmw} = \begin{cases} 2.69266 \times 10^{-10} (\lambda \leq 2) \\ 10^{-10} \cdot \exp \left[2416 \left(\frac{1}{303} - \frac{1}{T} \right) \right] \cdot [0.87 \cdot (3 - \lambda_{nf}) + 2.95 \cdot (\lambda_{nf} - 2)], (2 < \lambda_{nf} \leq 3) \\ 10^{-10} \cdot \exp \left[2416 \left(\frac{1}{303} - \frac{1}{T} \right) \right] \cdot [2.95 \cdot (4 - \lambda_{nf}) + 1.642454 \cdot (\lambda_{nf} - 3)], (3 < \lambda_{nf} \leq 4) \\ 10^{-10} \cdot \exp \left[2416 \left(\frac{1}{303} - \frac{1}{T} \right) \right] \cdot [2.563 - 0.33 \cdot \lambda_{nf} + 0.0264 \cdot \lambda_{nf}^2 - 0.000671 \cdot \lambda_{nf}^3], (\lambda_{nf} > 4) \end{cases} \quad (24)$$

$$D_{nmw} = \begin{cases} 3.1 \times 10^{-7} \lambda_{nf} [\exp(0.28 \lambda_{nf}) - 1] \exp \left(\frac{-2346}{T} \right), (0 < \lambda_{nf} < 3) \\ 4.17 \times 10^{-8} \lambda_{nf} [161 \exp(-\lambda_{nf}) + 1] \exp \left(\frac{-2346}{T} \right), (3 \leq \lambda_{nf} < 17) \end{cases} \quad (25)$$

4.1.2. Porous layers

Water transport and phase changes in porous layers (CL, MPL, and GDL) are factors critical to the PEM fuel cell cold start. This subsection discusses ionomer water diffusion in the CL and water phase changes in porous layers. Emphases are made on two subjects: the supercooling phenomenon and the effects of the MPL on transport.

4.1.2.1. Ionomer water transport in CL. In the catalyst layer, water exists in various states, such as ionomer water, liquid water, vapor, ice, and supercooled liquid water. Ionomer water transport in the CL is essentially the same as that in the membrane. The difference is that a part of the CL is composed with carbon backbones and can be regarded as an interconnected reticular structure. Hence the effective diffusion coefficient should be corrected by the transfer cross-section area.

4.1.2.2. General descriptions of water phase change in CL. Water phase change in the CL is more complex compared to that in the membrane due to structure porosity and ionomer/carbon mixing composition. As mentioned in Section 2, supercooled water is not likely to be present at very low temperatures. Therefore, a common assumption is that water in the CL exists in four states without supercooled water during cold start [5]. This theory suggests that there are two paths for the non-frozen membrane water to change to ice, as shown in Fig. 7. In the first path, non-frozen membrane water freezes to ice directly, under the condition that fully humidified gases are supplied to the CL. In the second path, when water vapor is not saturated in gases around the ionomer, non-frozen water first changes to vapor and then desublimates to ice. Evidently, the second path is more likely to occur during the cold start because non-humidified gases are commonly supplied in order to suppress the rate of ice formation. Many models are developed based on this theory, and the model predictions reasonably agree with experimental results, as previously discussed in Section 3.

4.1.2.3. Origin of the supercooling phenomenon. The ice formation theory without supercooled water cannot explain some behaviors of the cold start. An important one is the significant cycle variations. It was found that when the cold start is repeated many times, operation times before voltage failure could vary notably from cycle to cycle [29,116,117]. Oberholzer et al. [29] demonstrated that operation time variations are stochastic. As shown in Fig. 12, there is no correlation between the fuel cell operation time and high-frequency resistance (HFR) prior to the cold start. The operation time is not independent of maximum voltage, either. In summary, all observations indicated that the cold start cycle variation is a stochastic

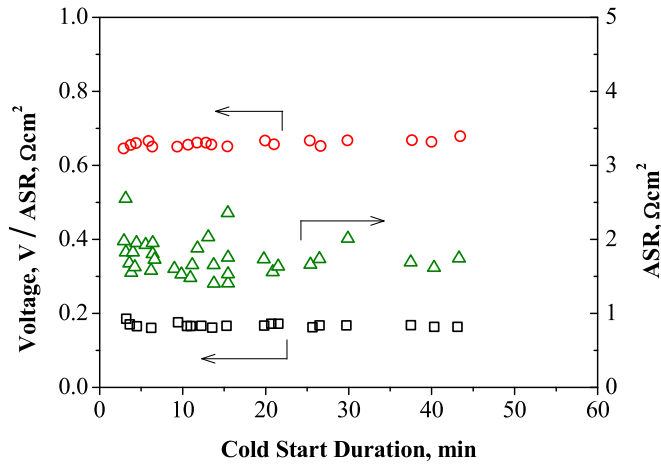


Fig. 12. Distribution of cold start durations over 34 cold start cycles (-10°C , 0.1 A cm^{-2}). Circles represent cell HFR prior to start, triangles represent cell HFR at maximum voltage, and squares represent maximum voltage (adapted from [29]).

process, rather than the result of cell degradation. Therefore, supercooling could be an explanation for these findings.

Many factors can affect the transition of supercooled water. For instance, existing research suggests that substrate surfaces with higher hydrophobicity provide better conditions to maintain supercooled water. Other environmental conditions, such as the humidity of surrounding gas, ambient flow velocity, and evaporative cooling, also affect the nucleation rate of supercooled water [67]. In principle, the liquid phase can be maintained until the temperature decreases to the crystal homogeneous nucleation temperature which is -48.3°C at standard pressure if no nucleus exists [68].

Real devices are of course more complex than idealized models. When water stays freezing temperatures in a liquid state, it can flow out of porous layers into the flow channel and is then taken away by the gas flow. In this situation, water can be drained efficiently unless the rate of water generation is very high; as a result, the fuel cell can work normally. However, supercooled liquid water can partially block the pores. Once nucleation is triggered, supercooled water suddenly freezes to ice, causing the water to block gas supply pathways. As a result, a sudden voltage failure occurs to the fuel cell. The seeds that trigger supercooled water release may be defects of fiber, cracks, or ice fragments [116]. They are randomly distributed in porous layers. Vibration can also act as a trigger, since vibration generates cavitation bubbles, and the collapse of bubbles causes a local pressure increase, leading to nucleation [118]. Therefore, the phase change trigger from supercooled liquid to ice could occur at any time.

Experiments have been conducted to visualize water and ice during cold start, and while some observed supercooled water, others did not. Ishikawa et al. [6] observed the supercooled water directly through micrography and found that given certain conditions, liquid water can remain supercooled and flow out of the CL and GDL. When the supercooled droplets grow larger than a critical value, they suddenly freeze. However, other experiments did not detect the presence of supercooled water. Ge and Wang [32] found that liquid water only emerged at a relatively high temperature (-3°C), since the temperature in the CL exceeded melting point soon after cold start. When the start temperature was lower than -3°C , only ice could be found on the CL surface at any given time. Although this experiment could not prove that supercooled water does not exist inside the CL, since only the CL surface was observed, the findings cast some doubt on the existence of supercooled water. The contradicting results from these two experiments also suggest the stochastic behavior of supercooled water.

4.1.2.4. Effects of cold-start factors on the supercooling phenomenon. After the direct observation of supercooled water, further experiments were conducted to analyze the correlation between the behavior of supercooled water and various factors during the cold start. This section summarizes four major factors that influence the behavior of supercooled water: initial temperature, initial water content, current density, and pore diameter. First, supercooled water is more likely to be present at a higher initial temperature [21,57]. Secondly, supercooled water is also more likely to exist at higher current densities [29,57]. A study showed that without supercooled water, water instantly froze in the CL and voltage slowly dropped [57]. When supercooled water was present, it flowed from the CL to GDL, which suddenly froze, leading to a sudden drop of voltage. Third, an initial dry condition favors the existence of supercooled water [6]. This may be due to the residual water that turns to ice before the cold start; the starting ice effectively acts as seeds, which promotes nucleation. Thus, the supercooled water is unstable without purging. Fourthly, smaller pore size favors supercooled water. A study on pore size found that supercooled water could be sustained inside the CL, which has a smaller pore size, but supercooled water flowing out of the CL is more likely to freeze at the interface between the CL and GDL [98]. Similarly, it was found that supercooled water was more likely to freeze inside the larger gaps between the CL and MPL, but when the gaps were small, supercooled water tended to flow through the gaps and into the MPL [119].

4.1.2.5. Models of supercooling phenomenon. Theoretical models of the supercooling phenomenon in PEM fuel cells can be found in [74,98]. A supercooled liquid cluster grows as water is generated and aggregates until a critical cluster radius is reached. The supercooled water is then released and ice is formed, which means that this spontaneous process leads the unstable supercooled water to a stable state [98]. The critical cluster radius is influenced by temperature, as shown in Eq. (26). The increase in supercooling causes a decrease in critical radius, as shown in Fig. 13. If the critical radius is on the same scale as the CL pore radius, supercooled water can be maintained. This corresponds to a supercooling temperature of -20°C .

$$r^* = -2\sigma_{cl} \frac{T_e}{\rho \cdot h \cdot \Delta T} \quad (26)$$

4.1.2.6. Suggestions on future supercooling modeling. Theoretical models for supercooling still need improvements and have yet to be submodeled for macroscopic cold-start models. For future modeling

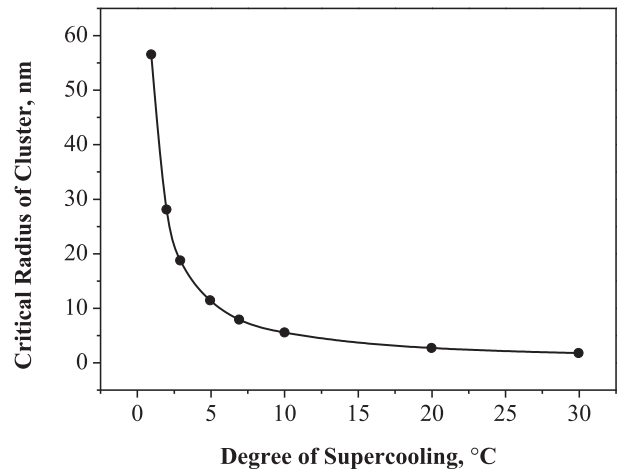


Fig. 13. Effect of temperature on critical cluster radius of supercooled water droplets (adapted from [98]).

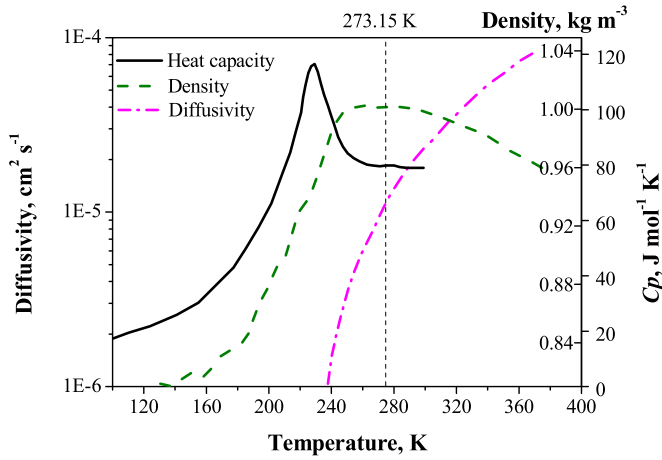


Fig. 14. Properties of supercooled water as a function of temperature (adapted from [120–122]).

work, two challenges should be considered. First, the properties of supercooled water are different from that of normal-state water. Fig. 14 shows the density [120], specific capacity [121], and diffusivity [122] of supercooled water over a range of temperatures. It is observed that these properties change significantly with temperature. Therefore, accurate property data are needed for different types of porous materials. Second, the presence of supercooled water is likely to be stochastic [116]. The existence of supercooled water is very sensitive to conditions; for instance, the transformation of supercooled water to ice can be triggered by a seed, such as fiber defects, cracks, or vibrations.

From a practical viewpoint, fuel cell designs should benefit from the presence of supercooled water to facilitate the cold start. Designs should also reduce cycle variations to improve the reliability of a successful cold start. For example, by increasing hydrophobic material loading in porous layers, seeds for ice nucleation could be covered and thus reducing the tendency to freeze [116].

4.1.2.7. Ice formation in GDL. Ice formation and melting kinetics in the GDL were studied [97,100,123]. The differential scanning calorimetry (DSC) method was adopted to experimentally determine the pseudo steady-state nucleation rates and crystallization rates of water in the GDL as functions of subcooling. The convolution integral over nucleation and growth rate is given as follows:

$$\phi = 1 - \exp\left(-\frac{4\pi}{3}g(\theta)\int_{\tau_i}^t J[T(t')]r^3(t-t')dt'\right) \quad (27)$$

where ϕ is the volume of water that transforms to ice, and t is time,

$$g(\theta) = \frac{1}{4}(2 + \cos\theta)(1 - \cos\theta)^2 \quad (28)$$

J is the nucleation rate given in the overall pseudo steady-state as

$$J = A \exp\left[-\frac{B}{T(\Delta T)^2}\right] \quad (29)$$

where A and B are constants.

4.1.2.8. The microporous layer. Microporous layers have been widely used in PEM fuel cells in both laboratory and industry settings [124]. Its effectiveness in improving voltage output in normal operating conditions has been confirmed. However, the effects of the MPL on cold-start performance require further discussions.

MPL can be made hydrophobic or hydrophilic. Hydrophobic MPL is commonly used to improve operational performance of PEM fuel cells. The mechanism is that hydrophobic MPL can increase water

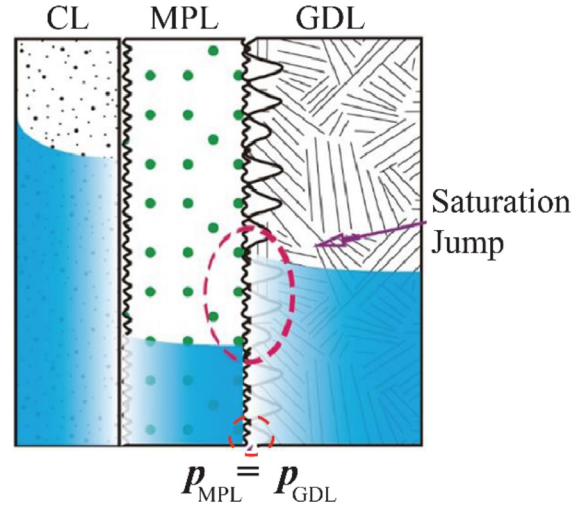


Fig. 15. Schematic diagram of liquid saturation jump at the interfaces between the catalyst layer (CL), microporous layer (MPL), and gas diffusion layer (GDL).

pressure and concentration on the outer side of the CL. Thus, more water flows into the membrane due to the higher concentration and pressure gradient. As a result, the membrane is better hydrated and has lower electrical resistance, and output performance is improved. MPL has a low saturation of liquid due to its hydrophobicity, as illustrated in Fig. 15. Therefore, it is also effective at preventing flooding, especially at a high current density and low temperatures.

4.1.2.9. Effects of MPL on liquid water transport. Before discussing the effects of MPL on the cold start, it is necessary to understand how MPL affects liquid water transport in porous layers. Based on the results of existing research, we can conclude that the MPL affects the cold-start process mainly by influencing the transport of supercooled water in its liquid state. The key to the effects of MPL is the structural change of the CL interfacing surface. Fig. 16 shows the SEM image of the through-plane cross-section of the cathode CL-MPL-GDL [66]. The three layers have different microstructures and various porosities. GDL is in a fiber form, and if the MPL is not inserted between the CL and GDL, there will be large spaces between the CL and GDL. The MPL is denser than the GDL, so it creates much smaller gaps between these surfaces.

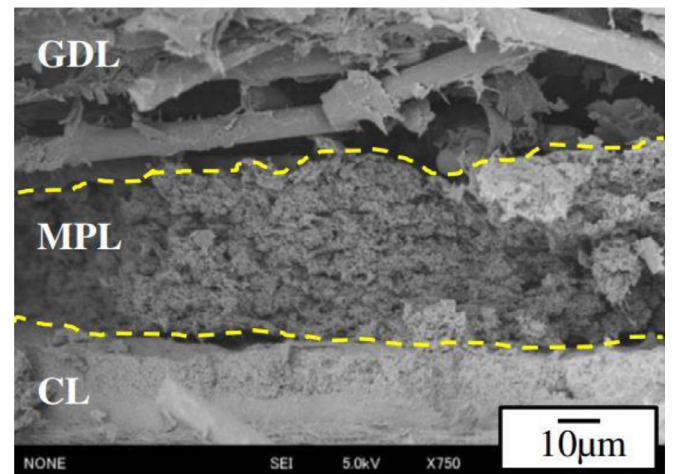


Fig. 16. SEM image of the through-plane cross section of the catalyst layer (CL), microporous layer (MPL), and gas diffusion layer (GDL) (reprinted from [66] with the permission of Elsevier).

The improvement of liquid transport with the MPL is not only critical for the cold start but also benefits operational performance. Many studies have observed the distribution of water in porous media during cell operation through high-resolution neutron imaging or X-ray radiography [125–128]. Discontinuous liquid saturation was identified across different layers. This appears to be due to the continuous capillary pressure in multilayers. Modeling studies clearly describe this phenomenon [129–131]. Liquid water transport in porous media is two-phase transport, and capillary pressure exists at the interface between liquid and gas. This is defined as the difference of liquid pressure and gas pressure:

$$p_c = p_{lq} - p_{gas} \quad (30)$$

The capillary pressure is usually given by the Leverett function [132]:

$$p_c = \begin{cases} \sigma \cos \theta \left(\frac{\varepsilon}{K_0} \right)^{0.5} \times [1.42(1-s_{lq}) - 2.12(1-s_{lq})^2 + 1.26(1-s_{lq})^3], & \theta < 90^\circ \\ \sigma \cos \theta \left(\frac{\varepsilon}{K_0} \right)^{0.5} \times [1.42s_{lq} - 2.12s_{lq}^2 + 1.26s_{lq}^3], & \theta > 90^\circ \end{cases} \quad (31)$$

In the capillary pressure equation, σ is surface tension:

$$\sigma = -0.0001676T + 0.1218(273.15K \leq T \leq 373.15K) \quad (32)$$

Capillary pressure is the driving force of liquid diffusion, and the liquid water diffusion coefficient is calculated as the following:

$$D_{lq} = -\frac{K_{lq}}{\mu_{lq}} \frac{dp_c}{ds_{lq}} \quad (33)$$

where K_{lq} is the effective permeability of water in the CL, which is calculated based on Eq. (34).

$$K_{lq} = K_{CL,GDL}^0 s_{lq}^4 (1-s_{ice})^4 \quad (34)$$

Because the different layers (CL, MPL, and GDL) connect as continuous porous media, the capillary pressure across them should also be continuous. This means that capillary pressure on each side of the interface will be equal, as they are both governed by the Leverett function:

$$p_c = \sigma \cos(\theta_{MPL}) \left(\frac{\varepsilon_{MPL}}{K_{MPL}} \right)^{0.5} J(s_{MPL}) = \sigma \cos(\theta_{GDL}) \left(\frac{\varepsilon_{GDL}}{K_{GDL}} \right)^{0.5} J(s_{GDL}) \quad (35)$$

Of course, porosity parameters are different for the two layers. Therefore, water saturation at the interface is supposed to experience a jump [131], as shown in the diagram in Fig. 15. The jump magnitude of saturation at the interface mainly depends on the material's hydrophobicity and pore radius of the two layers.

4.1.2.10. Effects of MPL on cold start performance. There are fewer studies focusing on the effects of MPL on the cold start, and these studies have not reached a consensus on the mechanisms [65,116,133]. Some studies found that both hydrophobic and hydrophilic MPLs can improve cold-start performance [133], while some studies asserted that hydrophobic MPLs had adverse effects [65,116]. Yet another study claimed that more ice was formed at the CL-MPL interface due to the hydrophobic MPL, preventing water from flowing inside MPL [65]. This situation reportedly deteriorated the cold-start performance.

After a comprehensive review of existing studies, we conclude that both hydrophobic and hydrophilic MPLs are capable of improving cold-start performance; their effects ultimately depend on other conditions, of which the initial temperature is most important.

The hydrophilic MPL improves cold start performance through simply extending the capacity of the CL, so more ice can be held

before the failure of the cold start. Thus, the cold start duration can be longer [65,134].

The mechanism of the hydrophobic MPL involves the presence of supercooled water. As previously discussed in above subsections, supercooled water is more likely to remain as liquid at higher temperatures and smaller droplet sizes. Therefore, the effects of hydrophobic MPLs on the cold start are highly dependent on initial temperature. When the temperature is relatively high, such as at -10°C , the supercooled liquid is sustained for longer. In this case, the MPL can force more water to flow into the membrane (the mechanism for this has been discussed above). Thus, the membrane becomes more conductive. More heat can be generated, and the temperature of the cell increases faster. This is supported by the finding that the anode MPL effectively improves cold-start performance [133]. Humidification in the anode can reduce resistance, while ice is not likely to block the anode. On the other hand, the interface of the CL-MPL is less rough than the CL-GDL interface, which better sustains supercooled water. If the initial temperature is too low, such as at -20°C , there is a lower chance of supercooled water flowing into the MPL, and ice is more likely to form inside the CL or on the CL surface. In this situation, the effect of MPL is less notable [21].

Fig. 17 illustrates the effects of MPL on the cold start process, from subfreezing to normal temperatures. We categorize the cold start process into three distinct stages during the evolution of temperature. The comparison between the cases with and without MPL assist in understanding how the MPL impacts transport and cold-start ability. The first stage covers the period before the temperature in MEA reaches melting point. When the ionomer is saturated, liquid water appears in a supercooled state and moves outside the CL. The supercooled liquid can be maintain while the radius of the water cluster is smaller than the critical cluster [98]. The supercooled water accumulates in the pores of porous media. With smaller pores, the cluster is less likely to reach that critical value, so the water can maintain its supercooled state and move outside the CL. Without MPL, the rough surface of the GDL contacts the CL to form large gaps. The supercooled water at this interface grows larger without confinement, finally reaching the critical radius and freezing into ice. With MPL, contact is finer at the interface between the MPL and CL, and more supercooled water can diffuse out of the CL into MPL. Thus, there is less ice formation on the surface of the CL allowing the cold start to last longer. Visualizations of ice on the CL surface after the cold start have confirmed the above discussions [21]. A recent study compared the rough and smooth interfaces of the CL-MPL, and the results also confirmed that the interface structure is the key to the behavior of supercooled water [119]. Thus, it is critical to cold-start performance.

The second stage is defined as when MEA temperature reaches the melting point and ice begins melting. In the case without MPL, the liquid formed can potentially cause the cathode to flood and result in performance deterioration due to the massive amount of ice existing on the surface of the CL [135]. Worse yet, due to the temperature gradient, the temperature of the GDL may be still lower than the melting point, so the thawed water from the CL may diffuse into the GDL and freeze again [25]. MPL can also mitigate such problems in this stage. For example, due to the existence of MPL, less ice accumulates on the surface of the CL in the first stage. In the second stage, the improved contact area between the MPL and CL benefits water draining and prevents CL flooding [21,66].

In the third stage, all the ice has been melted. Because the temperature is still low, the evaporation rate is low, and water draining is relatively inefficient before it reaches normal operation temperature. The MPL is also important at this stage, as it improves water draining.

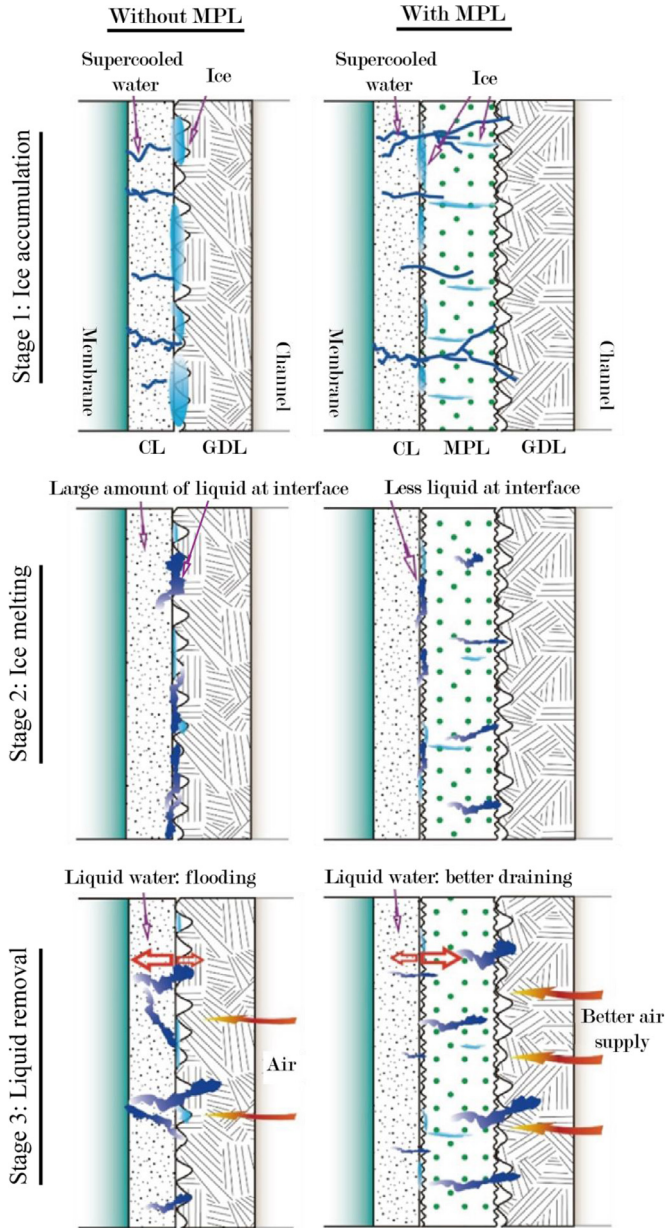


Fig. 17. Schematic diagram of the three-stage cold start process with the presence of supercooled water. Comparison of situations with and without the MPL.

4.1.3. Flow channels

Flow channels act as pathways for supplying reactants and draining water products. The primary transport process in the flow channel is convection due to its large size. As observed in an experiment, ice formation may occur in the flow channel when starting at a relatively high subfreezing temperature, such as -7°C [25]. The temperature of the CL and GDL exceeded melting point due to localized heat generation. Liquid water then flowed out of porous layers and into the flow channel. However, the temperature of the flow channel was still lower than the freezing point. As a result, water froze in the flow channel and blocked the pathway, leading to a high pressure drop and quick voltage failure, as shown in Fig. 18. To avoid the ice blockage in the flow channel, the interdigitated flow channel design resulted in better performance than the parallel channel [136]. The interdigitated channel enhanced water back-diffusion into the membrane. This design benefited membrane hydration and therefore resulted in higher conductivity. It also reduced the amount of water

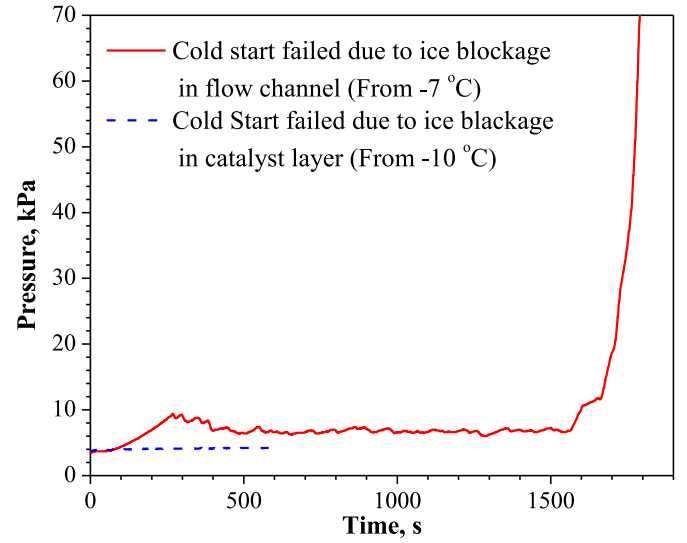


Fig. 18. Evolution of pressure drop in the cathode flow channel due to ice blockage (adapted from [25]).

in the flow channel and mitigated ice blockage. The effects of water and ice blockage on gas transport in the flow channel are discussed in the next section.

4.2. Gas species transport

4.2.1. Flow channels

As discussed above, ice formation in the cathode flow channel may cause a pressure drop in the airstream, which leads to a non-uniform current density distribution and possible start-up failure. In an unobstructed flow channel, the pressure drop is only caused by frictional loss [137], as a function given in Eq. (36):

$$\Delta P = \frac{1}{2} \rho V^2 \left(\frac{4fL}{D_h} + \sum_{i=1}^n \xi_i \right) \quad (36)$$

where f is the friction factor, ξ_i is the bend loss coefficient of the i th bend. In cases of blockage, additional pressure loss may occur. The blockage effect of a droplet is associated with droplet size and contact angle.

4.2.2. Porous layers

During a cold start, ice and liquid in porous layers can prevent reactants from contact with the catalysts, also resulting in voltage failure. Ice and liquid block reactants from flowing into the CL. Ice in the CL also covers the surface, thus reducing the electrochemical active area (ECA).

Gas species transport in the CL should consider the Knudsen diffusion because the pore diameter of the CL tends to be very small and of the order of the mean free path [99]. The Knudsen diffusion can be given in the form of Fick's law:

$$N_i^K = -D_i^K \nabla c_i \quad (37)$$

where D_i^K is the Knudsen diffusion coefficient, as given in Eq. (38):

$$D_i^K = \zeta^K \cdot r_{CL} \sqrt{\frac{8RT}{\pi M_i}} \quad (38)$$

Considering ice and liquid blockage in the CL, the effective diffusivity can be calculated as

$$D_{CL,i}^{eff} = D_i^K \varepsilon^{\tau_d} (1 - s_{lq} - s_{ice})^{\tau_d} \quad (39)$$

In the GDL, Knudsen diffusion can be neglected, so the effects of liquid and ice blockage on gas species transport in porous media can be given as Eq. (40).

$$D_{GDL,i}^{eff} = D_i \varepsilon^{\tau_d} (1 - s_{lq} - s_{ice})^{\tau_d} \quad (40)$$

The effect of ECA reduction on performance loss can be formulated by a modified Butler–Volmer equation for the anode and cathode, respectively.

$$j_a = (1 - s_{lq} - s_{ice})^{\tau_a} j_{0,a}^{ref} \left(\frac{C_{H_2}}{C_{H_2}^{ref}} \right)^{0.5} \times \left[\exp\left(\frac{2\alpha_a F}{RT} \eta_{act}\right) - \exp\left(-\frac{2\alpha_c F}{RT} \eta_{act}\right) \right] \quad (41)$$

$$j_c = (1 - s_{lq} - s_{ice})^{\tau_c} j_{0,c}^{ref} \frac{C_{O_2}}{C_{O_2}^{ref}} \times \left[\exp\left(\frac{4\alpha_a F}{RT} \eta_{act}\right) - \exp\left(-\frac{4\alpha_c F}{RT} \eta_{act}\right) \right] \quad (42)$$

In Eqs. (39) and (40), τ_d is the ice blockage coefficient for gas diffusivity reduction. In Eqs. (41) and (42), τ_a is the ice coverage coefficient for ECA reduction. These two parameters should be determined based on the morphology of the ice formed in the CL. However, it is challenging to measure ice morphology in the CL, since the formation characteristics are sensitive to cold-start conditions and ice formation processes show random behaviors. In models, these values are defined by assumptions. Some studies used 1 for τ_a , 1.5 for τ_d , [5,77], while others used 0.5 for τ_a and 1.5 for τ_d [94]. Mishler et al. [30] found that an ice fraction-dependent τ_a can produce better agreement with experiments. Clearly, a better understanding of the supercooling phenomena would lead to an improved understanding of ice blockage.

4.3. Heat transfer and heat sources

The thermal conductivity of the membrane is usually assumed to be a constant in PEM fuel cell modeling studies, although it has been found that thermal conductivity varies notably depending on water content. Burheim et al. [138] examined the dependency of the thermal conductivity of Nafion™ as a function of water content. The results indicated that the thermal conductivity of the Nafion™ could be enhanced by increasing water content, as shown in Eq. (43).

$$k_{Nafion} = (0.177 \pm 0.008) + (3.7 \pm 0.6) 10^{-3} \lambda \quad (43)$$

The effects of temperature were explored in another study [71], and the thermal conductivity of the humidified membrane is governed by Eq. (44).

$$k = \frac{(\lambda_w / \rho_w) k_w + (1 / c_{SO_3} - M_w) k_{nf}}{1 / c_{SO_3} - M_w + \lambda_w / \rho_w} \quad (44)$$

where λ_w is the water content, ρ_w (kg m⁻³) is the water density, k_w (W K⁻¹ m⁻¹) is the water thermal conductivity, c_{SO_3} (mol m⁻³) is the molar concentration of the SO₃⁻ ion, and k_{nf} (W K⁻¹ m⁻¹) is the thermal conductivity of the dry membrane. The thermal conductivity of water increases with temperature, whereas the situation is reversed for a dry membrane. Therefore, the humidified membrane's thermal conductivity is essentially determined by the two effects combined.

During a cold start, the rising temperature in the CL determines the performance of a successful start-up. The thermal resistance of the CL is obtained by measuring the MEA resistance and membrane resistance. It depends on porosity, ionomer content, and platinum load. It is also influenced by installation pressure [71]. While the catalyst layer is relatively thin in the through-plane direction, low thermal resistance causes a low temperature gradient. A more important factor is the heating source.

The reaction occurs in the catalyst layer. Therefore, heat from the electrochemical reaction is a major heating source from the CL, and it can be calculated as activation heat and reversible heat. Ohmic heat is another major source of heat, especially when the water content in ionomer is low (creating high resistance) or the operational current is high. Latent heat from water phase changes (evaporation and ice formation) also contributes to the CL. All the above-mentioned four heating sources are compiled in Eq. (45).

$$\dot{Q}_{gen,total} = \int_V \left[\underbrace{j \eta_{act}}_{\text{Activation heat}} + \underbrace{\frac{j_c T \Delta S}{2F}}_{\text{Reversible heat}} + \underbrace{\|\nabla \phi_{ele}\|^2 \kappa_{ele}^{eff} + \|\nabla \phi_{ion}\|^2 \kappa_{ion}^{eff}}_{\text{Ohmic heat}} + \underbrace{\sum \left(\frac{\dot{m}_{pc} \dot{h}_{pc}}{VC} \right)}_{\text{Latent heat}} \right] dV \quad (45)$$

4.4. Charge transport

The proton transport process depends on water content and water state, and water state corresponds to temperature, as discussed above [5]. When membrane water freezes, the ionic conductivity of the membrane drops sharply.

Charge transport in the CL involves the electron transport in carbon/Pt and proton transport in ionomer. Due to the porous structure and composition of ionomer and carbon, the effective charge transport coefficients can be calculated based on the volume of the conductive medium. Different approximations can be used to correct the charge transport coefficients, such as the Bruggeman approximation, Looyenga approximation, and generalized Maxwell equation [139].

4.5. Summary

In this section, the transport phenomena of the cold start, as based on various experimental observations and theoretical studies, were discussed. Water transport in the membrane is influenced by water content and temperature, and the freezing ability of membrane water is dominated by the bonding of sulfonic groups and molecular membrane structures. Water transport in porous layers is a key factor in cold-start performance. Supercooled water is more likely to emerge when initial temperature is high, current density is high, and the initial state is dry. In ideal conditions, supercooled water remains in the CL, where pores are small. When supercooled water flows out from the CL, it accumulates in the gaps between CL and GDL. Liquid droplets grow and then suddenly freeze when their size exceeds a critical value. In real conditions, the nucleation of supercooled water is a stochastic behavior, which can be triggered by randomly distributed cracks, fiber defects, and ice fragments. Hydrophobic MPLs can improve cold-start performance in certain temperature ranges by forcing more water to flow into the membrane and preventing ice formation on the CL surface. Hydrophilic MPLs can also improve cold-start performance by extending the ice capacity of the CL. To obtain more precise simulation results, future cold-start modeling should consider the behavior of supercooled water and the effect of MPLs.

5. Optimizations

Fundamental studies on the PEM fuel cell cold start have provided insight into the key factors that influence the ability of the start-up. For practical purposes, these key factors can be informative for cold-start optimization, so that more critical cold-start performance requirements can be achieved for FCV commercialization.

The improvement of cold-start performance is not merely an issue of rapid heating. It should be examined in conjunction with fuel economy, system stability, and operation simplification. In this section, we categorize the optimization routes for the cold start into three aspects: design and material parameters, starting load control strategies, and assisted cold start with external energy.

5.1. Design optimization

PEM fuel cell cold-start performance can be optimized through better-designed parameters and choice of materials. Studies suggest that the optimization of critical cell components (such as the membrane, CL, MPL, GDL, and BP) and efficient designs at the stack and system level can improve cold-start performance. For a summary of research results and suggestions on design parameter optimization, see Table S3 in Supplementary Materials.

5.1.1. Membrane

As shown in Table S3, optimization of the membrane focuses on water-uptake ability and conductivity at subfreezing temperatures. The idea is simple: if more water can be absorbed in the membrane, less ice will be formed and the start-up will be more likely to succeed. One particular method of achieving this is to reduce the thickness of the membrane [104]. They analyzed three types of NafionTM with different thicknesses (NafionTM 117, 115 and 112). It was found that the thinner membrane (NafionTM 112) is most beneficial to the cold start. The water concentration gradient of the thinner membrane is greater than that of the thicker membranes. This results in a higher rate of water absorption, so ice accumulates at a slower rate with the thinner membrane. The water capacity of the membrane can also be improved by using advanced materials [140]. The pure NafionTM membrane was compared to a composite membrane of SiO₂ and Zr(HPO₄)₂ nanoparticles. Experimental results showed that the composite membrane contained more free water, allowing for better performance of the cold start.

Low temperatures can lead to reduced conductivity of the membrane, causing serious voltage degradation. Some studies claimed that there was no noticeable phase transition in the membrane during the cold start, based on in-situ experiments, since temperature increased significantly in the membrane [43]. Nevertheless, there are experiments that observed a sudden change in ion conductivity of the membrane when temperature decreased, suggesting a possible freeze in the membrane [148]. The freezing of membrane water was found to be correlated to membrane nanostructures [33,108] and chemical composition [33]. The pore diameter of the membrane significantly affects the depression of the freezing point. As for the nature of the membrane, two types of membranes (NafionTM and sulfonated polyimide) were compared [33]. Special behaviors were observed for the hydrocarbon-based membrane: water did not freeze upon cooling and also did not desorb upon annealing. These behaviors are explained by the strong affinity between the novel membrane material and water. This characteristic is favorable to the cold start by preventing water diffusion to the catalyst layer and ice formation in the pores. However, there are concerns about the stability of this type of membrane; more research should be proposed to examine its durability in freezing conditions.

5.1.2. Catalyst layer

In the catalyst layer, critical factors affecting the cold start include ionomer content, Pt/carbon ratio, composition, and thickness. By increasing the fraction of ionomer in the CL, the rate of ice accumulation can be effectively reduced, since the higher ionomer fraction can absorb more water and drive more water to diffuse into the membrane [104]. Hiramitsu et al. also experimentally confirmed the advantages of higher ionomer fractions, and they reasoned that

the benefits come from better oxygen permeation in the ice-blocked catalyst layer [17].

Electrode thickness also affects the water capacity of the membrane. Thompson et al. [45] found that the maximum water content in the membrane for a thick electrode (14.1 μm) is around 14, while for a thin electrode (3.5 μm), the maximum water content is around 15. Although a cold start favors a thick catalyst layer, this increases precious metal loading and may deteriorate durability. An ultrathin CL is prone to flooding, especially at a lower initial temperature [141]. To remedy the performance loss caused by the ultrathin CL, some strategies were proposed, such as using a thicker membrane and drying the membrane before start-up.

The effects of the platinum-to-carbon support ratio in the catalyst layer cathode were also studied [142]. It was found that a lower platinum/carbon support ratio was more likely to reduce the rate of ice formation. As for the composition of CL materials, it was found that the addition of silica in the cathode catalyst layer could improve water capacity and therefore improve the time sustained during the cold start [143].

5.1.3. Microporous layer

Microporous layers can be added between the CL and GDL to improve the transport process. Studies have found that the MPL can improve cold-start performance, as it expands ice capacity in the CL [134,144] and reduces water accumulation on the surface of the CL [21]. A dual-functionality MPL was designed for both normal operation and cold start [134,144]. In normal operation conditions, the MPL exhibits better water management, as it improves the water draining ability of the GDL [149]. In the cold start, its major function is to expand the ice capacity of CL, therefore reducing the risk of ice blockage in the CL. Another study also found that MPL improves cold-start performance by reducing ice accumulation on the CL surface [21]. This is through the better contact between the MPL and CL, which benefits water removal from the surface.

On the other hand, freeze damage to the CL resulting from ice expansion may be intensified by the MPL, because water is constrained within smaller spaces at the interface of the CL and MPL. When ice is formed at the interface between the CL and MPL, the resulting pressure could be higher than in the absence of the MPL, and hence more severe CL pinhole damage or micro-cracking could occur. This could delaminate the catalyst layer from the GDL [145,150].

5.1.4. Gas diffusion layer

As for the GDL, its pore size and thickness are two key factors influencing cold-start performance. Hirakata et al. [76] studied the effects of GDL pore size. They found that a GDL with smaller pores is favorable to the cold start, as it worked longer at a constant current at −5 °C. They further analyzed the effects of ice formation on the reaction area and found that the reaction area of the catalyst layer was maintained even after ice formation and apparent cell potential decrement. The study showed that voltage failure is mainly due to the loss of reactant gas transport rather than the blockage of reaction sites. They also found that the GDL has no effect on the cold start at temperatures below −10 °C because water would freeze before it could flow into the GDL. Dursch et al. [97] investigated the characteristics of ice melting in the GDL. They formulated an equation (Eq. (46)), which related GDL thickness (L) and thermal conductivity (k). This equation suggests that a thinner GDL could accelerate ice melting and therefore is favorable to the cold start.

$$t_{\text{melt}} = \sqrt{\left(\frac{\varepsilon \rho_{\text{liq}} S_{\text{liq}} H_f}{\beta}\right) \left(\frac{\sigma_{\text{GDL}}^2}{k_{\text{eff}}} + \frac{2\sigma_{\text{GDL}}}{h_0}\right)} \quad (46)$$

5.1.5. Flow channels

Interdigitated, parallel, and serpentine patterns are commonly used PEM fuel cell flow channel designs. Interdigitated flow-field is favored over the parallel design in the constant current start-up mode, since the interdigitated flow-field enhances water removal from under-land GDL through convective mass transport [136]. At a lower start-up current density, the fuel cell with an interdigitated flow channel successfully started up from $-6\text{ }^{\circ}\text{C}$, while the parallel flow channel fuel cell could only sustain $-4\text{ }^{\circ}\text{C}$ under the same conditions. As for the effects of flow direction, both experimental and modeling studies have found that the influence either of co-flow or counter-flow are negligible because the cathode flow dominates other cold-start characteristics [25,89].

5.1.6. Stack

In PEM fuel cell stack design, two factors primarily affect cold start ability: the material and structure of bipolar plates. Metallic materials have many advantages when applied to bipolar plates. In terms of the cold start, metallic bipolar plates can reduce the thermal mass of the stack, thus speeding up the cold-start process [93]. However, metallic bipolar plates are more likely to suffer from corrosion and high mechanical stress during the cold start [146]. The performance of metallic bipolar plates can be improved by depositing vanadium oxide polycrystalline [147]. The negative temperature coefficient feature of the film is favorable for the cold start, since it becomes a self-heating source in the cold start process. As for the endplates, high thermal conductivity causes an increase in start-up time due to the fact that the high conductivity contributes to a higher heat dissipation rate to the cold ambient [93].

5.2. Load control

Hardware provides foundations for cold-start performance optimization, and operating strategies, such as load control, can further enhance the cold start in more convenient and efficient ways.

5.2.1. Current

The constant-current cold start mode is used in many experimental and modeling studies. Since current is easily controlled, the constant-current mode is advantageous for stabilizing flow rates in experiments. Optimizing current density generally encompasses two approaches. One is based on the heat utilized before a shutdown occurring after full ice blockage [43,44]. In this case, a lower start-up current density favors a low rate of water generation, allowing more time for the ionomer to uptake water and more time for the accumulation of ice in the CL pores. Although the heat generation rate would be lower compared to a high current density start-up, the total heating time would be longer, and as a result, more overall heat could be utilized. The other perspective focuses on the competition between rising temperature and ice accumulation [104]. It is reasoned that the higher final temperature at the end of the failed cold start suggests the higher possibility of a successful startup. Based on this view, a higher current density would be favorable because of the higher heat generation rate for the same amount of water produced.

Another method for improving the cold start is current-ramping [103]. This strategy was developed based on earlier findings. Research suggests that lower operational current density slows down ice formation rate, while higher current density contributes to a rapid warm-up. Linear current-ramping start-up combines both of these advantages. At the initial low current, ice is prevented from forming; as the temperature rises and the operational current increases, since the water capacity has already been increased, ice formation remains at a low rate, even as more water is produced.

5.2.2. Voltage

Although the constant-voltage mode is harder to control, studies have shown it to be superior to the constant-current mode [5,86,104]. Low-voltage operation can maintain a high rate of heat generation during the entire cold-start process [104]. Compared to the constant-current start mode, the constant-voltage start-up can produce more waste heat and hence enables the cell to reach a higher final temperature. Faster heat-up can be achieved when a lower voltage is applied [87]. Therefore, the PEM fuel cell is more likely to survive a cold start at a lower voltage. In practical applications, such as FCVs, low-voltage operation is commonly applied during the cold-start period. Unfortunately, this compromises output power and efficiency to achieve rapid heating. The constant voltage start-up mode can be created by regulating the current with a DC-DC converter [151–153]. Generally, the method of converting entails raising voltage through a boost converter, at the cost of reducing current density, or by reducing voltage through a buck converter.

5.2.3. Power

In practical application, fuel cells are usually favored to operate with a constant load to obtain optimum performance and efficiency, such as when fuel cells are used as a part of a hybrid system in vehicles [154,155]. The power-controlled cold start modes will be discussed in this subsection. In particular, the constant power mode and maximum power mode will be analyzed.

The characteristics of cold-start failure in the constant power mode are shown in Fig. 19 [156]. In this case, a 20-cell stack started up from $-20\text{ }^{\circ}\text{C}$ with a constant power of 80 W. As shown in Fig. 19 (a), due to the increase in membrane resistance, current density decreased sharply. A defining characteristic of the constant power start-up is the high starting current density. The high current causes a sudden redistribution of membrane water in the MEA through the electro-osmotic drag effect. Thus, there is a sharp decrease of water content in the anode CL. As a result, the total resistance of the cell increases and current density sharply drops in the first few seconds. Fig. 19(b) and (c) show the evolution of the polarization curves displayed in the power–current density (P–I) pattern and the voltage–current density (V–I) pattern, respectively. The polarization curves move downward in this figure, which indicates the degradation of performance during the cold start. The peak value of the lowest curve is the same as the operational power (80 W), and this point indicates the moment of failure for the constant-power cold start. Even though the pores of the cathode CL were not fully obstructed by ice at this point as shown in Fig. 19(a), the starting process terminated because the required power output could not be maintained after this moment.

In addition to constant-power operation, the fuel cell can be started with maximum power output to quickly reach full load. In a failed maximum-power cold start, power output drops sharply due to a large ohmic loss, caused by fast dehydration on the anode side of MEA (driven by EOD effect). It then decreases steadily until complete failure. If the temperature in the MEA reaches melting point, ice in the CL will melt and drain, which would improve operating conditions. Power output rises and tends to stabilize as the ice melts completely and liquid drainage becomes steady. The maximum power mode was found to be the most survivable cold-start mode, especially when the fuel cell started from an extremely low temperature [157]. The advantages of the maximum power mode can be explained by the heat generation characteristics, as shown in Fig. 20. As opposed to the constant-voltage mode in which the heat generation rate drops in later stages, heat generation increases constantly after the first drop stage in the maximum-power cold start mode. This advantage is critical for the rapid warm-up.

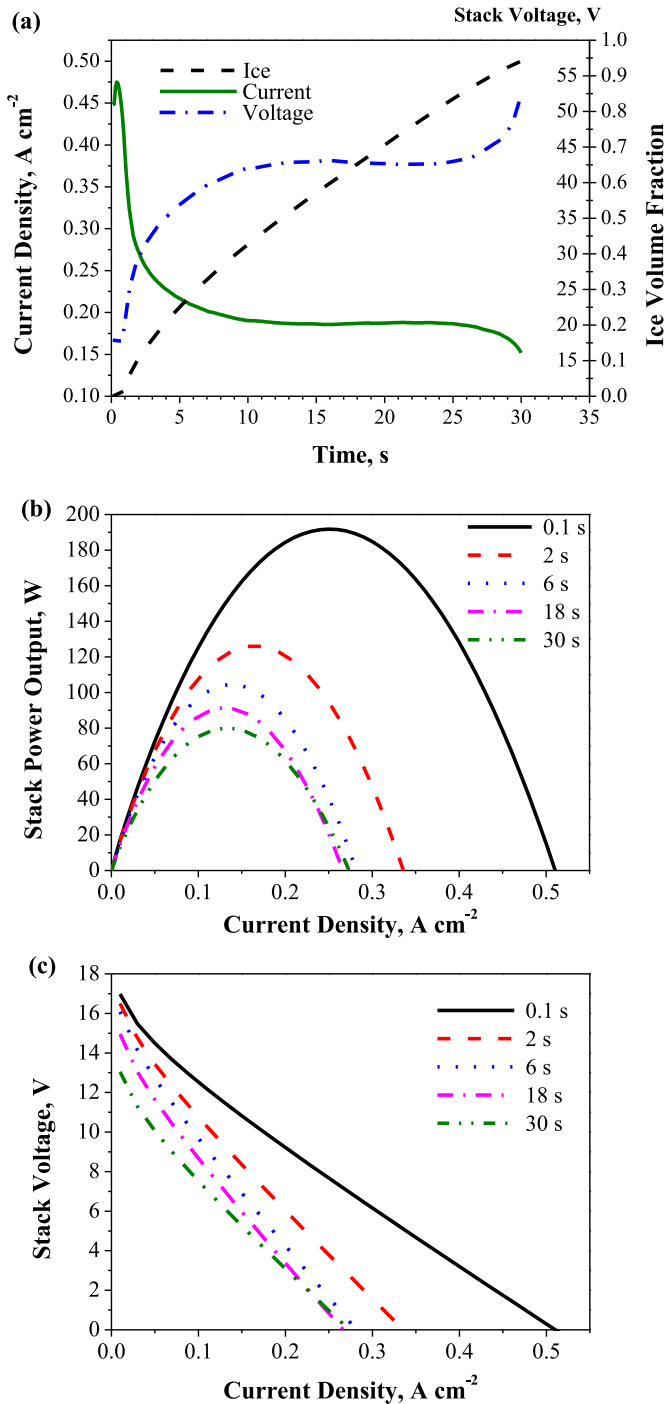


Fig. 19. Stack performance predictions at a high current density, constant power mode. Operation power is set to 80 W. (a): evolution of current density, stack voltage, and average ice volume fraction in cathode catalyst layers; (b): evolution of power current density curves; (c): evolution of voltage-current density curves (adapted from [156]).

5.3. Assisted start-up

This paper defines the assisted cold start as a positive cold-start method with auxiliary equipment or methods in addition to the essential components in a fuel cell system. Optimizations of starting modes are not included. As revealed in fundamental research, the key factor to a successful cold start is the ratio of cell warm-up time constant to ice accumulation rate [77]. Therefore, purging and heating are two essential aspects of the assisted cold start. Various

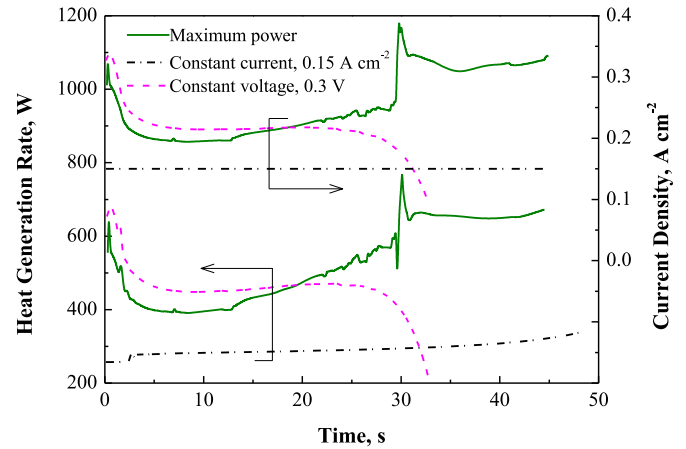


Fig. 20. Evolution of total heat generation rate and current density for three different cold start modes: maximum power, constant current at $0.15\ A\ cm^{-2}$, and constant voltage at 0.3 V. The initial temperature is $-12.5\ ^\circ C$ for all cases (adapted from [157]).

assisted cold start methods are summarized in Table S4 [16,92,94,104,158–170] in Supplementary Materials.

5.3.1. Purging

Several studies have found that conditions prior to a cold start are critical to a successful cold start; these conditions especially include initial water content in the membrane and catalyst layer [79, 87]. The purge process has been widely studied through experimental [158,161,171–173] and modeling [173–175] studies.

The purge process can be described in four stages, as shown in Fig. 21 [174,175]. First, the purge gas flows through the flow channel and blows away water droplets on the channel walls. There are several experimental [176–179] and modeling studies [180–183] that focus on this two-phase flow problem. Second, liquid water in the GDL pores is removed by evaporation and the driving force comes from capillary pressure [184,185]. Afterwards, the liquid water in the CL pores is removed. Finally, the membrane water in ionomer evaporates. As the amount of liquid water in the CL pores is reduced, the surface area of ionomer exposed to gas increases, which further accelerates the removal of membrane water.

There are several indicators that reflect the level of purge effectiveness. The first is cell resistance [161,172]. Since the resistance of the membrane and CLs is strongly affected by water content, the efficiency of drying can be evaluated by measuring cell resistance. The second is the dew point temperature of the purge gas at exit, which indicates the amount of water contained in the gas [161]. Thirdly, the weight of MEA also shows the extent of drying. Visualizing the GDL and CL is another direct method to observe and quantify the effectiveness of water removal through purging [161]. Two mass transport parameters that determine purge performance were found [172]. One is the diffusivity in the CL and GDL. The other is the convection of vapor in the gas flow channel. This is confirmed by comparing the purge effects between N_2 and helium. Because the water vapor diffusivity of helium is much higher than that of N_2 , a helium purge results in better performance.

Practical applications of PEM fuel cells require high purge effectiveness and a short purge duration. The conventional purging method still needs further development. Several novel purging strategies can be found in literature, as shown in Table S4 in Supplementary Materials. Tang et al. [158] experimentally studied a vacuum-assisted purge method, which was first presented by Fuss [159] in a patent. Nitrogen is used as the purging gas, which minimizes the degradation of the CL. The vacuum-assisted method has high purge efficiency while consuming less purging gas, an important advantage for on-board applications. The HFR test shows that high-pressure,

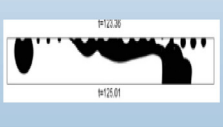
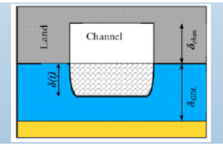
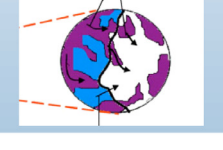
Sequence	Water state	Location	Transport	Schematic
Stage 1	Liquid drop removal	Flow channel	Convection	
Stage 2	Liquid water	GDL	Evaporation & Capillary pressure driving	
Stage 3	Liquid water	CL pores	Evaporation & Capillary pressure driving	
Stage 4	Membrane water	CL ionomer & Membrane	Evaporation & Diffusion	

Fig. 21. Sequence of the gas purge process and corresponding transport phenomena.

vacuum-assisted drying is the most effective method among the tested cases. However, the mechanical stress on the MEA would be higher with high pressure gradients, so the pressure should be limited. An experiment was conducted that introduced a small amount of hydrogen into the cathode air-purge flow [161]. The catalytic hydrogen-oxygen reaction in the CL improved the ability of the purge to remove water, especially compared to a purge using only air. This was very effective at drying the catalyst layer. Subsequent cold start tests after using the new purge strategies also exhibited better performance.

Although gas purging can effectively remove residual water in the PEM fuel cell and improve cold-start performance, it may cause degradation problems in catalyst layers. Furthermore, the extra energy consumption and increase in cold-start time also must be considered for practical applications [174]. Future development of rapid purge methods can focus on inflicting less damage to the catalyst layers.

5.3.2. Heating

In automotive applications, the extra heat provided to fuel cells for starting up in subfreezing environments can be categorized into two types: the warm-parking method and the positive heating method [7]. The warm-parking method requires a continuous supply of energy and is not optimal for long parking times. Some studies have presented strategies to improve warm-keeping performance, such as using phase-change materials [96] or using porous water-transport plates to slow down ice formation after shutting down in freezing conditions [186].

Here, discussion will focus on the positive heating method. Extra heat can be provided from various energy sources. It is important to analyze and compare the effectiveness, availability, and energy consumption level of these heating sources.

5.3.2.1. Electric heating. The major advantage of electric heating is the flexibility in control, which benefits the relatively complicated assisted cold-start strategies. Many proposed patents for assisted cold-start methods are based on electric heating, as shown in Table S4 in Supplementary Materials [164,187]. Zhou et al. presented a novel variable heating and load-control assisted cold-start method using an electric heater as the heating source [94]. As shown in Fig. 22, a 100 W electric heating source is distributed along a 20-cell stack. The inner cells are usually started first, and the waste heat is used to heat up the adjacent cells, which allows a successful cold start of the stack at -20°C .

5.3.2.2. Gas heating. Other assisted fuel cell cold-start patents are based on the gas heating method [16,188,189]. The effects of inlet temperature on cold-start performance were studied [5,16,93,104]. One such study compared two inlet gas conditions. The inlet gas temperatures were either set equal to the ambient temperature or equal to the cell temperature; based on the evolution of current density, the results showed negligible differences in cold-start performance [5]. In this simulation, the inlet gas temperature that was equal to the cell temperature was not much higher than the ambient temperature. More notable effects on cold-start performance were found by further increasing the inlet gas temperature [104]. The inlet gas temperatures of -20°C and 80°C were compared. At the higher inlet gas temperature, current density dropped slower. However, the start-up still failed even when the inlet gas temperature was at 80°C . Two caveats about these results should be mentioned. Firstly, this model only considered a straight channel in a cell; therefore, compared to a full-size single cell or stack, the heat carried by the high temperature inlet gas would not be effectively utilized before the gas flowed out of the straight channel. Secondly, only the cathode side was heated up; due to the lower heat transfer coefficient of air compared to hydrogen, heating the cathode side is much less efficient than heating the anode side [93]. More reasonable results of

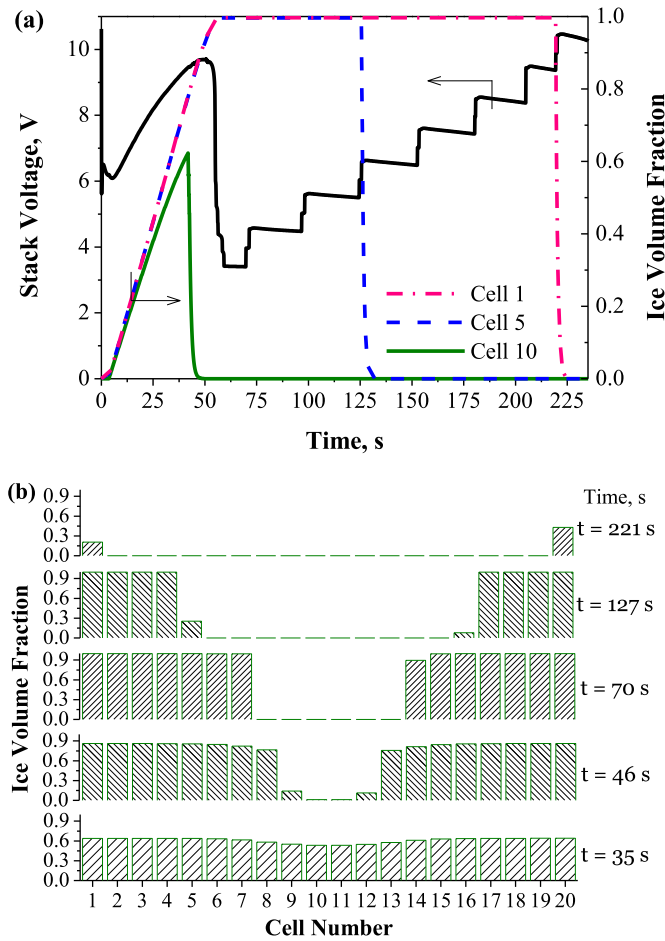


Fig. 22. Cold start characteristics of a 20-cell stack using the external heating assisted-start method. (a): stack voltage and ice volume fraction in cathode catalyst layers; (b): ice volume fraction in cathode catalyst layers in different cells of a 20-cell stack, starting from -20°C at 0.15 A cm^{-2} . The two end surfaces are thermally insulated, and an external heating power of 100 W is evenly applied to the 6 middle cells (adapted from [94]).

the effect of inlet gas temperature can be found in [16]. The experiment showed that a cold start from -10°C failed when the inlet gas temperature was at 60°C and succeeded when the inlet gas was at 80°C .

5.3.2.3. Coolant heating. There are several patents using the coolant heating method for the assisted cold start [170,190–192]. Some studies have mentioned that the stack can warm up quickly by passing through hot coolant during the cold start [92,93,95]. One advantage of this method is that in a practical situation, hot coolant can be obtained from various sources. Hot coolant can come from electricity consumption (for example, in a hybrid vehicle with fuel cells and an internal combustion engine) as well as from fuel cell stack coolant, which can be heated by the engine's waste heat. Another benefit of this method is that the warm-up process can be uniform and efficient, as the coolant channels are evenly distributed inside the stack. It also avoids having to install extra accessories, a downside of the direct electric heating method.

A problem with this method is the relatively high heat capacity of coolant, which could lead to higher energy consumption. This issue may slow down the cold start process, considering that the coolant, itself a large thermal mass, needs more time to heat up. Some cold start research recommends the deactivation of coolant loops during the cold start [24].

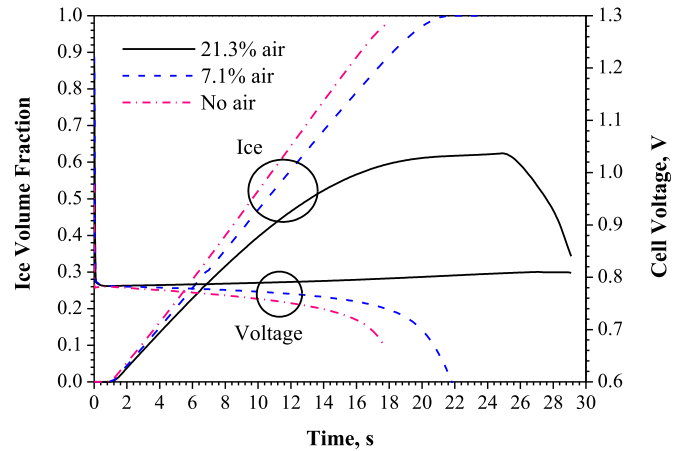
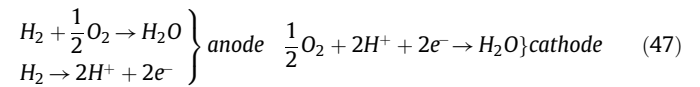


Fig. 23. Evolutions of volume-averaged ice volume fraction in the cathode catalyst layer and cell voltage under constant-current conditions with 21.3%, 7.1%, and 0% air in the anode inlet gas mixture. The current density is at 0.1 A cm^{-2} (adapted from [167]).

5.3.2.4. Internal catalytic hydrogen-oxygen reaction. In 1823, it was found by Döbereiner that hydrogen and oxygen can react slowly on a platinum surface at room temperature [193]. The fundamental mechanism has been explained in detail [194]. The application of the catalytic reaction to assist the PEM cold start is straightforward [195–197]. Several patents on this method have been made [166,198–202]. The catalytic reaction can occur at low temperatures and the reaction gases are readily available from the reactant supply system, making it suitable for PEM fuel cells.

Two schemes can be applied to the internal hydrogen-oxygen catalytic reaction to assist the cold start. These are based on whether or not the catalytic reaction activated simultaneously with the start-up. In other words, the first type of scheme is the application of the catalytic reaction on the catalyst layer before the fuel cell starts working. The fuel cell is initiated when the cell is warmed up to a proper condition; for instance, an average temperature above 0°C [165]. The second type is to simultaneously apply the catalytic reaction with the cold-start process [167]. The reactions in anode and cathode CLs are as follows:



As shown in Fig. 23, if the air-to-hydrogen ratio is high enough (within the explosion limit), a cold start at -20°C could be successful. When the temperature of MEA exceeds melting point before the cathode CL is fully blocked, the ice melts and leads to a successful cold start (Fig. 24).

5.3.2.4. External burner. The external burning of the hydrogen-oxygen method can be found in multiple patents [203–206]. It is a high-efficiency method that rapidly heats up the fuel cell stack, since the controlled catalytic burning of a hydrogen-oxygen mixture is already well developed [207–212]. Even with these advantages, this method is not practically feasible for automobile applications due to the increased weight and volume of the extra burner. Adding complexities to the installation and control of auxiliary components also presents further problems.

5.4. Summary

In this section, the optimization strategies for improving PEM fuel cell cold-start performance are discussed. Optimization methods are categorized into three types: improving the structure and material of cell components, optimizing operation modes, and

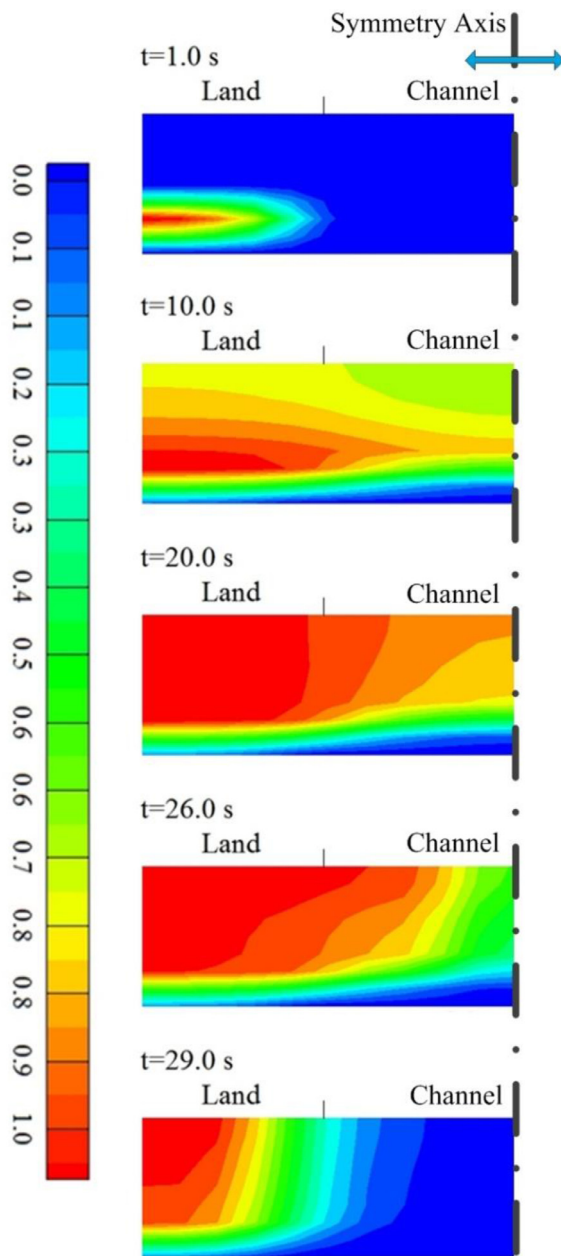


Fig. 24. Transient distribution of ice volume fraction in the middle cross-section of the cathode catalyst layer with a constant-current cold start with 21.3% air mole fractions in the gas mixture at the anode inlet (adapted from [167]).

providing extra energy to effectively remove residual water and to achieve rapid heating. For the first type, structural design optimization, it is critical to maintain free water content and the conductivity of the membrane. For the catalyst layer, water absorption is essential for improving cold-start performance. This favors bipolar plates with lower heat capacity. In the second type, operation modes for the cold start include current control, voltage control, and power control. The constant-power and constant-voltage modes are better at improving cold-start performance, compared to the constant-current mode. Although the assisted cold start tends to be more effective, the feasibility and reliability of the various methods need to be considered, especially for applications in compact products (such as the FCV), which require lighter and smaller fuel cell systems. Applying an assisted cold start method to a PEM fuel cell system is more complex than the mere addition of system components; the interactions between the subsystems must also be considered.

6. Challenges and opportunities

6.1. Existing challenges and expected improvements

The major challenges of the PEM fuel cell cold start can be categorized into three aspects. The first is the understanding of the water phase transition phenomenon. The second is the modeling technique, which can balance both microscopic modeling and large-scale system level modeling. The third is the gap between cold-start performance requirements and existing technologies.

6.1.1. Liquid–ice transition theory

Ice formation from liquid water occurs commonly during the cold start, but its nature and characteristics are still unclear, given the complex conditions of the cold start process. In literature, the primary level of cold start studies ignored the presence of ice. These studies mainly focused on thermal management. Later, advanced second-level studies captured the ice formation process with empirical equations, either in lumped models or multiphase numerical models. Although these studies were able to demonstrate the impact of ice formation on the cold start process, their empirical descriptions have proven to be inadequate. For example, empirical models cannot explain stochastic performance variations [29] and the presence of supercooled water [65] during the cold start. Third-level studies focused on the supercooling phenomenon and have been able to improve theories on ice formation [6,98]. Two issues still remain with regards to liquid–ice transition research. The first is that this theory is based on several hypotheses that cannot be validated by experiments thus far. The second issue is that these theoretical models are limited to simply describing the supercooling phenomenon itself. The supercooling theories have not yet been integrated into a complete cold-start model for the full-scale fuel cell.

Improvements in the ice formation theory require detailed experimental observations. Therefore, experimental techniques must be improved with the goal of observing dynamic nano-scale transport phenomena in the cold start. As reviewed in Section 2, existing experiments provide some unique techniques to observe water at the nano-scale in PEM fuel cells, such as CRYO-SEM [61]. The CRYO-SEM method can visualize nano-scale water distribution in the CL, but due to its static nature, it is unable to capture the liquid–ice transition process. Some other techniques, such as neutron imaging, can capture the dynamics of ice formation, but observations currently cannot be made in nano-scale measurements; the resolution of the neutron imaging technique is at the 10 μm level [54,55].

6.1.2. Multi-scale modeling

Multi-scale modeling strikes a balance between the microscopic level and large-scale system level models and are useful for future studies of the cold start. It is still difficult to achieve, primarily due to the limited understanding of microscopic phenomena and its high computational cost.

Although pore-scale modeling has been achieved for water transport in the GDL and CL, consideration of the ice formation mechanism at the pore-scale level is still limited within the scope of theoretical analysis [59,61,109]. There are no numerical models for pore-scale simulation of the cold start process. One of the major challenges is the lack of understanding of the ice formation mechanism at this level, although there are already exist some studies on ice formation characteristics in a confined space [213,214] and at the nano-scale [215]. Ice formation in the catalyst layer of PEM fuel cell is especially complex due to the nature of the catalyst layer, which is composed of various materials. The pore size distributions are stochastic, and the properties of the interface are difficult to precisely control. Therefore, to alleviate these challenges, improvements could be made to develop a better theory regarding the ice formation mechanism in the cold start, as mentioned above.

Computational costs tend to be high for a variety of reasons. The grid or lattice size must be small enough to capture the features of pores, since pore size in the CL varies from the nano-level to the micro-level. Additionally, since the cold start process involves many transports and chemical reactions, numerous equations must be solved simultaneously. To reduce computational costs, pore-scale modeling is presently limited to the fuel cell scale. Code optimization should be considered for improving computational efficiency [216].

6.1.3. Gap between required performance and existing technologies

For cold start issues, a practical aim is improving the cold start ability without reducing its durability. Although there are many existing strategies for cold-start performance optimization, as reviewed in Section 5, there are some gaps in technical knowledge. Start-up time is one of these problems. Most of the existing assisted cold-start strategies require a relatively long time for heating so that cold starts can be successful when starting at extremely low temperatures. However, in practical applications, such as FCVs, start-up times need to be short enough for practical convenience. Another challenge is the compromise between performance and durability. Typically, the rapid cold start requires a much higher current density than in normal operation for the purposes of obtaining more waste heat [217]. Therefore, cold start operation tends to accelerate the degradation of electrodes. The trade-off between cold start ability and durability also exists on a material level. Optimal cold-start performance requires higher porosity of the catalyst layers, which allows them to withstand more ice and obtain a larger active area that promotes electrochemical reactions. However, these features generally result in a more unstable catalyst layer [218], thus leading to lower durability.

6.2. Opportunities

6.2.1. Experimental studies

Based on reviews of existing studies and analyses of cold-start challenges, four trends emerge in possible opportunities for future cold start research. One such field is the improvement of experimental techniques. Sophisticated experimental techniques, such as nano-scale observation [219,220], should be explored, enabling the observation of transport phenomena during the cold start process.

6.2.2. Modeling studies

Because critical water phase changes and mass transfers both occur at the interface of electrolytes, catalysts, and carbon backbone in the CL, it is important to comprehensively understand the interface phenomena with pore-scale modeling, especially for ice formation characteristics. So far, models for water transport in pores include the volume-of-fluid (VOF) model [185], pore-network model [221], and lattice Boltzmann model [222, 223], though perhaps only the lattice Boltzmann model is appropriate for the small pores in the catalyst layer. Future efforts can be made to model water phase change and ice morphology in the GDL and MPL. Importantly, these modeling techniques can be applied to the CL with electrochemical reactions.

6.2.3. Model uncertainty

Analyzing model uncertainty is another important aspect of cold-start modeling studies. For the PEM fuel cell cold start model, aleatory uncertainties exist due to the stochastic behavior of physical phenomena, such as the presence and release of supercooled water. Epistemic uncertainties also exist due to the difficulty in accurately measuring material properties, such as conductivity and reactivity of the catalyst layer. Future modeling studies should pay attention to model uncertainties. Model validation can be done by system-level experiments which measure cold-start performance, as well as

by component-level experiments, which can validate material properties and fundamental theories.

6.2.4. Material development

Material development is a possible major approach for further optimization of cold-start performance. Emphasis can be made in particular on the nano-structure design of electrode materials, as well as graphene-based electrode materials. The trade-off between cold start ability and durability is one of the major issues that can be resolved through material development. Future studies can focus on promoting the coupling of catalyst particles and graphene-based materials to improve both cold-start performance and durability [218].

7. Concluding remarks

Two important aspects of the cold start in proton exchange membrane (PEM) fuel cells involve transport phenomena and startup performance optimization. Fundamental studies through experiments and mathematical modeling are first reviewed. Transport phenomena during the cold-start process are then systematically discussed. The transport phenomena are summarized based on existing findings in various areas of research, such as output performance measurement, water and ice visualization, and theoretical model predictions. Next, cold-start performance optimization strategies are discussed, spanning topics such as enhanced material and structure, proper load control, and auxiliary system assistance. Finally, the challenges and opportunities facing cold-start research are discussed. Some important conclusions are addressed as follows:

1. The deterioration of fuel cell output performance during the cold start is strongly affected by water transport and ice formation processes, such as the presence of supercooled liquid water, ice formation in porous layers, and ice blockage of the flow channel. Therefore, a better understanding of the mechanism resulting in cold-start failure requires experimental methods, such as X-ray tomographic microscopy and CRYO-SEM, to reveal more about water transport and phase change.
2. Cold start models are developed at different levels of complexity with different methods. Each type of model suits a certain research purpose. Analytical cold start models often simplify fuel cell dimensions to zero in order to focus on interactions between key processes during the cold start, such as water balance, energy balance, and output performance. Analytical models are efficient tools for studying fundamental theories about the cold start. Numerical models can be implemented in various dimensions. They provide comprehensive details about the cold start process in full-scale cells. Thus, numerical models can be used to guide cell structure and material design for optimizing cold-start performance.
3. Supercooling is one of the most important transport phenomena in the cold start. Correlations between supercooled water and cold-start characteristics remain debated. Existing experimental findings show that supercooled water tends to stay in the liquid state in small pores of the catalyst layer. When it flows out and accumulates in larger gaps, such as the interface between the catalyst layer and the gas diffusion layer (GDL), this supercooled liquid water is more likely to freeze as droplets grow in size. The nucleation of supercooled water can be triggered by cracks, ice fragments, and defects of fiber, which are randomly distributed. Therefore, the freezing of supercooled water is likely a stochastic process. Understanding the supercooling phenomenon may assist in explaining some interesting behaviors in the cold start, such as the cycle variation of output performance during repeated cold starts.

4. The latest fuel cell designs commonly utilize microporous layers (MPLs) to improve cold-start performance. For hydrophilic MPLs, cold start performance is improved though increasing the ice capacity of the catalyst layer, thus prolonging the duration of the cold start. Hydrophobic MPLs enhance the cold start ability in certain temperature ranges by forcing more water into the membrane and preventing ice formation on the surface of the catalyst layer.
5. A key requirement for improving the cold start is optimal material and structural design of cells and stacks. For example, optimized ionomer content in catalyst layers, proper membrane thickness, and lower heat capacity materials can contribute to performance. Optimization of the start mode and load control, combined with the assisted heating method, may result in further improvements. Although using the assisted cold-start methods could be more effective, their feasibility and reliability need to be considered, especially when applied to compact products like fuel cell vehicles (FCVs).

Future cold start modeling studies should include and place emphasis on the supercooling phenomenon and water transport/phase-change in multi-layer porous media. Meanwhile, it is imperative to utilize more sophisticated experimental methods, such as real-time water and ice visualization and cryogenic microscopy, to validate models and theories; these further research directions will lead the field towards ultimate solutions to cold-start issues.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.pecs.2017.10.003.

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